```
%root_path = fileparts(mfilename("fullpath"));
%root_path = pwd;

% retrieve selected cuts used for further fitting
[HN_FFCutsFits,CutFitFileHN] =
recover_selected_cuts('HN_FFCutsFits','e:\SHARE\Fe\Data\analysis\07_fit_with_multic
ut_allJ\sel_cuts');
```

```
fn = fieldnames(HN_FFCutsFits);
nf = 11
```

```
nf =
11
```

```
%for i = 11:numel(fn)
%selectCuts = fn{nf};
selectCuts = 'Ei800_off1H1H0_HNminFF'
```

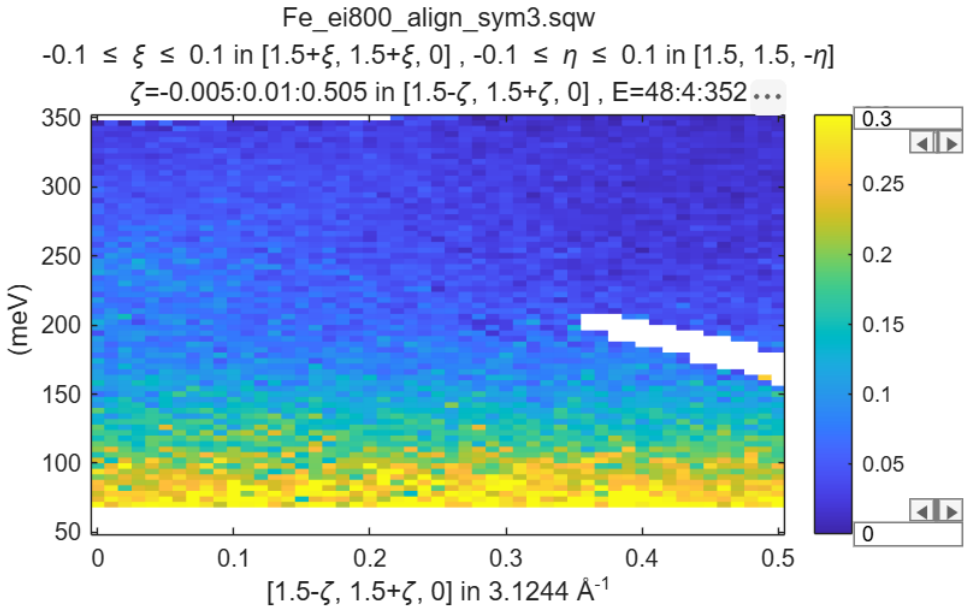
```
selectCuts =
'Ei800_off1H1H0_HNminFF'
```

```
if ismember(selectCuts(1:3),{'msk','dat'})||...
        contains(selectCuts,'all_fit_par')
    %continue
    error('Can not fit field: %s',selectCuts);
end

fprintf('ProcessiHN cuts: %s \n', selectCuts);
```

```
ProcessiHN cuts: Ei800_off1H1H0_HNminFF
```

```
cuts_list = HN_FFCutsFits.(selectCuts);
if ~iscell(cuts_list)
    cuts_list = {cuts_list};
end
if contains(selectCuts,'Ei200')
    min_Ei = 30;
    mi = maps_instrument(200,600,'S');
end
if contains(selectCuts,'Ei400')
    min_Ei = 40;
    mi = maps_instrument(401,600,'S');
end
if contains(selectCuts,'Ei800')
    min_Ei = 70;
    mi = maps_instrument(786,600,'S');
end
sample=IX_sample(true,[1,0,0],[0,1,0],'cuboid',[0.04,0.03,0.02]);
for j=1:numel(cuts_list)
    cuts_list{j} = cuts_list{j}.set_instrument(mi);
    cuts_list{j} = cuts_list{j}.set_sample(sample);
    plot(cuts_list{j}); lz 0 0.3; keep_figure;
end
```



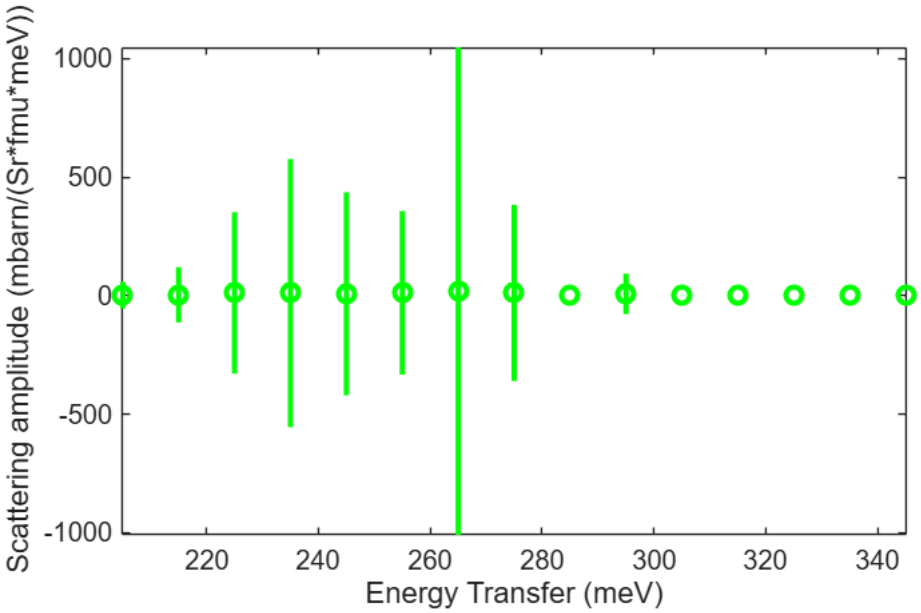

```
dir_name  = "HN";

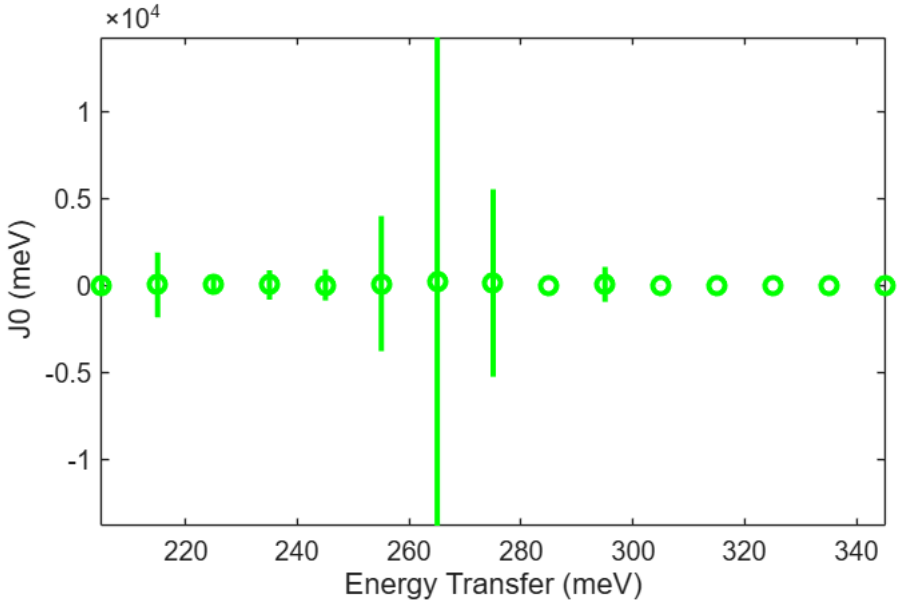
dE_step = 2; %original energy transfer step data were binned to. No point in going
finer
half_dE = 5; % half width of data binning

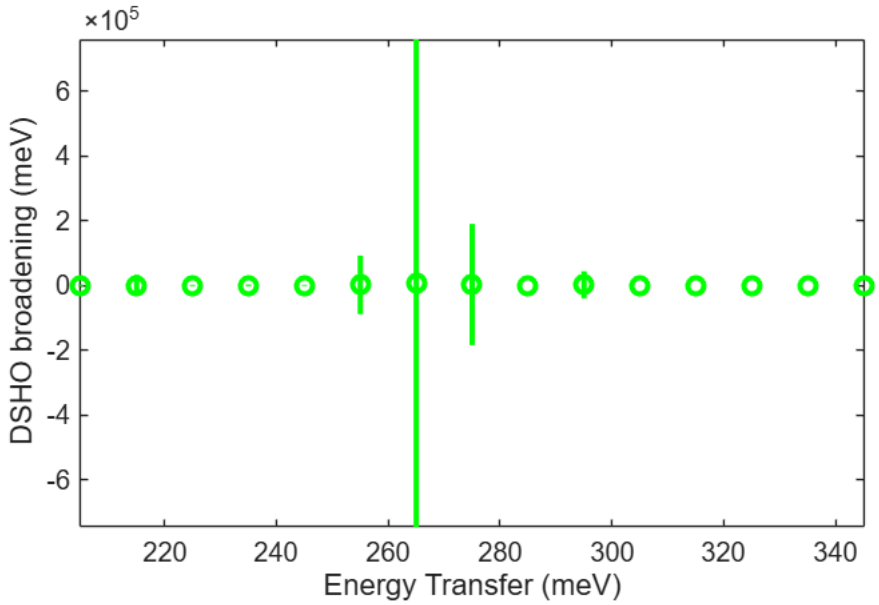
en_mm = cuts_list{1}.data.img_range(:,4);
en_mm(1) = max(min_Ei,en_mm(1));
en_mm(1) = 200;
%en_mm(2) = 380;
cut_en = en_mm(1)+half_dE:2*half_dE:(en_mm(2)-half_dE);

cut_dimensions = 1;
HN_FFCutsFits.(selectCuts) = cuts_list;
```

```
HN_FFCutsFits = fit_cut2JFF_along_direction(...
    HN_FFCutsFits,selectCuts,cut_dimensions,cut_en,dE_step,half_dE,true);
```


Scattering amplitude (mbarn/(Sr*fmu*meV))
1000
500
0
-500
-1000
220
240
260
280
300
320
340
Energy Transfer (meV)


×10^4
1
0.5
0
-0.5
-1
J0 (meV)
220
240
260
280
300
320
340
Energy Transfer (meV)




```
*****************************
**** En = 205±5 going to 345
*****************************
-------------------------------------------------------------------------------
Beginning fit (max 20 iterations)---------------------------------------------------------------------------------------------
Starting point
------------------
 Total time = 0.57411s    Reduced Chi^2 = 33.1393
 Free parameter values:
         172.9          2.158          48.27             35        0.05925
-------------------------------------------------------------------------------
Iteration = 1
------------------
 Total time = 2.0099s    Reduced Chi^2 = 1.194      Levenberg-Marquardt = 0.1
 Free parameter values:
         131.2          2.539          51.62             35        0.04251
-------------------------------------------------------------------------------
Iteration = 2
------------------
 Total time = 3.4s    Reduced Chi^2 = 28.7453      Levenberg-Marquardt = 0.01
 Free parameter values:
           201         0.1924           39.8             35        0.03923
 Total time = 3.6892s    Reduced Chi^2 = 1.0024      Levenberg-Marquardt = 0.1
 Free parameter values:
         159.8          1.897          47.28             35        0.04003
-------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 5.1267s    Reduced Chi^2 = 0.93472      Levenberg-Marquardt = 0.01
 Free parameter values:
         217.8          3.021          57.25             35        0.03858
-------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 6.5348s    Reduced Chi^2 = 2.325      Levenberg-Marquardt = 0.001
 Free parameter values:
         430.7          7.817          98.25             35        0.03768
 Total time = 6.8353s    Reduced Chi^2 = 0.87055      Levenberg-Marquardt = 0.01
 Free parameter values:
         295.6          4.549          70.02             35        0.03824
-------------------------------------------------------------------------------
Iteration = 5
```

```
------------------
 Total time = 8.3165s    Reduced Chi^2 = 2.3394      Levenberg-Marquardt = 0.001
 Free parameter values:
        629.3           11.7            126             35        0.03755
 Total time = 8.6158s    Reduced Chi^2 = 0.86943      Levenberg-Marquardt = 0.01
 Free parameter values:
        411.3          6.807          87.58             35        0.03803
--------------------------------------------------------------------------------
Iteration = 6
------------------
 Total time = 10.1291s    Reduced Chi^2 = 1.8341      Levenberg-Marquardt = 0.001
 Free parameter values:
        853.3          15.29          149.8             35        0.03765
 Total time = 10.4215s    Reduced Chi^2 = 0.85523      Levenberg-Marquardt = 0.01
 Free parameter values:
          560          9.502            107             35        0.03796
--------------------------------------------------------------------------------
Iteration = 7
------------------
 Total time = 11.9178s    Reduced Chi^2 = 2.2255      Levenberg-Marquardt = 0.001
 Free parameter values:
         1330          21.78          196.3             35        0.03752
 Total time = 12.2552s    Reduced Chi^2 = 0.86822      Levenberg-Marquardt = 0.01
 Free parameter values:
          814          13.39            135             35        0.03783
 Total time = 12.5287s    Reduced Chi^2 = 0.80307      Levenberg-Marquardt = 0.1
 Free parameter values:
        645.4           10.8          115.6             35        0.03796
--------------------------------------------------------------------------------
Iteration = 8
------------------
 Total time = 13.9975s    Reduced Chi^2 = 0.84999      Levenberg-Marquardt = 0.01
 Free parameter values:
        924.2             15          145.7             35        0.03785
 Total time = 14.3001s    Reduced Chi^2 = 0.81149      Levenberg-Marquardt = 0.1
 Free parameter values:
        734.8          12.14          125.1             35        0.03793
 Total time = 14.6351s    Reduced Chi^2 = 0.80158      Levenberg-Marquardt = 1
 Free parameter values:
        674.5          11.24          118.5             35        0.03797
--------------------------------------------------------------------------------
Iteration = 9
------------------
 Total time = 16.0767s    Reduced Chi^2 = 0.8124      Levenberg-Marquardt = 0.1
 Free parameter values:
          766           12.6          128.2             35        0.03791
 Total time = 16.3771s    Reduced Chi^2 = 0.79391      Levenberg-Marquardt = 1
 Free parameter values:
        703.7          11.67          121.6             35        0.03795
--------------------------------------------------------------------------------
Iteration = 10
------------------
 Total time = 17.8866s    Reduced Chi^2 = 0.8162      Levenberg-Marquardt = 0.1
 Free parameter values:
        803.4          13.11          131.8             35        0.03789
 Total time = 18.1744s    Reduced Chi^2 = 0.81174      Levenberg-Marquardt = 1
 Free parameter values:
        735.6          12.13          124.8             35        0.03793
 Total time = 18.4995s    Reduced Chi^2 = 0.80398      Levenberg-Marquardt = 10
 Free parameter values:
        713.9          11.82          122.6             35        0.03794
 Total time = 18.8148s    Reduced Chi^2 = 0.8066      Levenberg-Marquardt = 1000
 Free parameter values:
        704.8          11.68          121.7             35        0.03795
```

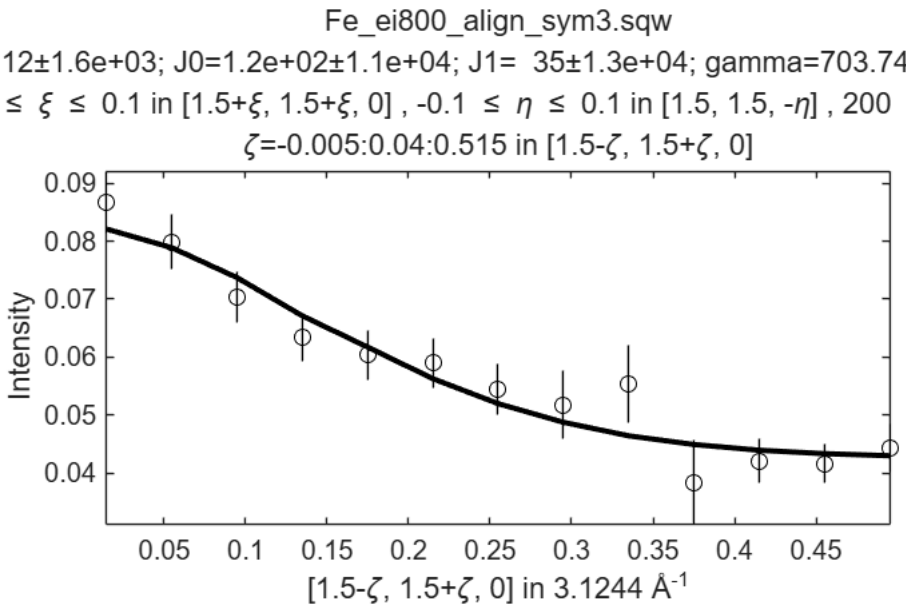
```
 Total time = 19.1046s    Reduced Chi^2 = 0.79714      Levenberg-Marquardt = 100000
 Free parameter values:
        703.9          11.67          121.6             35        0.03795
 Total time = 19.4148s    Reduced Chi^2 = 0.80162      Levenberg-Marquardt = 10000000
 Free parameter values:
        703.8          11.67          121.6             35        0.03795
 *** No improvement in chi-squared over previous iteration ***
 Total time = 19.436s    Reduced Chi^2 = 0.79391
 Free parameter values:
        703.7          11.67          121.6             35        0.03795
------------------------------------------------------------------------------
Fit converged:

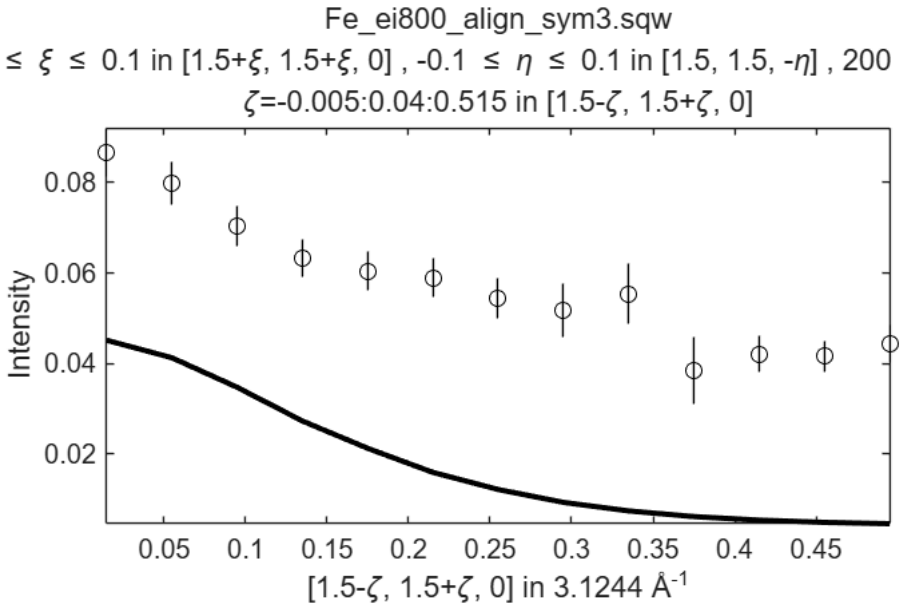
Parameter values (free parameters with error estimates):
---------------------------------------------------------

Foreground parameter values:
----------------------------
    1              1
    2              8
    3          703.7   +/-   9.852e+04
    4          11.67   +/-   1647
    5              0
    6          121.6   +/-   1.109e+04
    7             35   +/-   1.301e+04
    8              0
    9              0
   10              0


Background parameter values:
----------------------------
    1        0.03795   +/-   0.01975
    2              0
    3              0


Covariance matrix for free parameters:
--------------------------------------
    1.0000    0.9996    0.9999   -0.9999   -0.9822
    0.9996    1.0000    0.9999   -0.9999   -0.9774
    0.9999    0.9999    1.0000   -1.0000   -0.9795
   -0.9999   -0.9999   -1.0000    1.0000    0.9797
   -0.9822   -0.9774   -0.9795    0.9797    1.0000
```






```
*****************************
**** En = 215±5 going to 345
*****************************
-------------------------------------------------------------------------------
Beginning fit (max 20 iterations)---------------------------------------------------------------------------------------
Starting point
------------------
 Total time = 0.34785s    Reduced Chi^2 = 42.9158
 Free parameter values:
         692.8          3.567          76.22             35        0.05588
-------------------------------------------------------------------------------
Iteration = 1
------------------
 Total time = 1.9614s    Reduced Chi^2 = 2.4691      Levenberg-Marquardt = 0.1
 Free parameter values:
         357.7          3.399          79.75             35        0.04176
-------------------------------------------------------------------------------
Iteration = 2
------------------
 Total time = 3.5552s    Reduced Chi^2 = 2.0295      Levenberg-Marquardt = 0.01
 Free parameter values:
```

```
         769.7          4.728          99.88             35        0.03719
--------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 5.1713s    Reduced Chi^2 = 1.6143      Levenberg-Marquardt = 0.001
 Free parameter values:
          1328          4.952          108.3             35        0.03528
--------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 6.803s    Reduced Chi^2 = 1.1404      Levenberg-Marquardt = 0.0001
 Free parameter values:
         787.1          4.312          84.64             35        0.03491
--------------------------------------------------------------------------------
Iteration = 5
------------------
 Total time = 8.4556s    Reduced Chi^2 = 1.2662      Levenberg-Marquardt = 1e-05
 Free parameter values:
          1678          7.304          132.7             35        0.03465
 Total time = 8.7963s    Reduced Chi^2 = 1.1266      Levenberg-Marquardt = 0.0001
 Free parameter values:
          1110          5.191          100.7             35        0.03491
--------------------------------------------------------------------------------
Iteration = 6
------------------
 Total time = 10.4444s    Reduced Chi^2 = 1.1308      Levenberg-Marquardt = 1e-05
 Free parameter values:
          1724          6.773          127.1             35        0.03478
 Total time = 10.7516s    Reduced Chi^2 = 1.1125      Levenberg-Marquardt = 0.0001
 Free parameter values:
          1343          5.653          109.2             35        0.03487
--------------------------------------------------------------------------------
Iteration = 7
------------------
 Total time = 12.3039s    Reduced Chi^2 = 1.2876      Levenberg-Marquardt = 1e-05
 Free parameter values:
          2778          9.231          168.1             35         0.0346
 Total time = 12.67s    Reduced Chi^2 = 1.1181      Levenberg-Marquardt = 0.0001
 Free parameter values:
          1818          6.788            128             35        0.03475
 Total time = 12.9818s    Reduced Chi^2 = 1.11      Levenberg-Marquardt = 0.001
 Free parameter values:
          1509          6.004          115.1             35         0.0348
--------------------------------------------------------------------------------
Iteration = 8
------------------
 Total time = 14.6047s    Reduced Chi^2 = 1.1294      Levenberg-Marquardt = 0.0001
 Free parameter values:
          1129          5.099            100             35        0.03485
 Total time = 14.9589s    Reduced Chi^2 = 1.123      Levenberg-Marquardt = 0.001
 Free parameter values:
          1391          5.717          110.3             35        0.03482
 Total time = 15.2781s    Reduced Chi^2 = 1.1224      Levenberg-Marquardt = 0.01
 Free parameter values:
          1474          5.913          113.5             35        0.03481
 Total time = 15.6244s    Reduced Chi^2 = 1.1217      Levenberg-Marquardt = 1
 Free parameter values:
          1507          5.996          114.9             35        0.03481
 Total time = 15.947s    Reduced Chi^2 = 1.1261      Levenberg-Marquardt = 100
 Free parameter values:
          1509          6.003            115             35         0.0348
 Total time = 16.2714s    Reduced Chi^2 = 1.1206      Levenberg-Marquardt = 10000
 Free parameter values:
          1509          6.004          115.1             35         0.0348
```

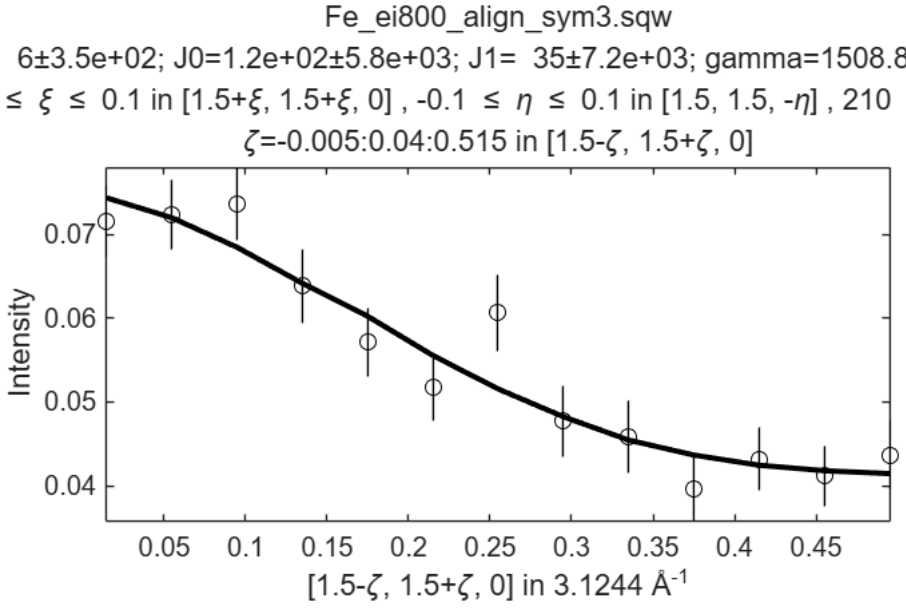
```
 *** No improvement in chi-squared over previous iteration ***
 Total time = 16.2805s    Reduced Chi^2 = 1.11
 Free parameter values:
          1509          6.004          115.1             35         0.0348
------------------------------------------------------------------------------
Fit converged:

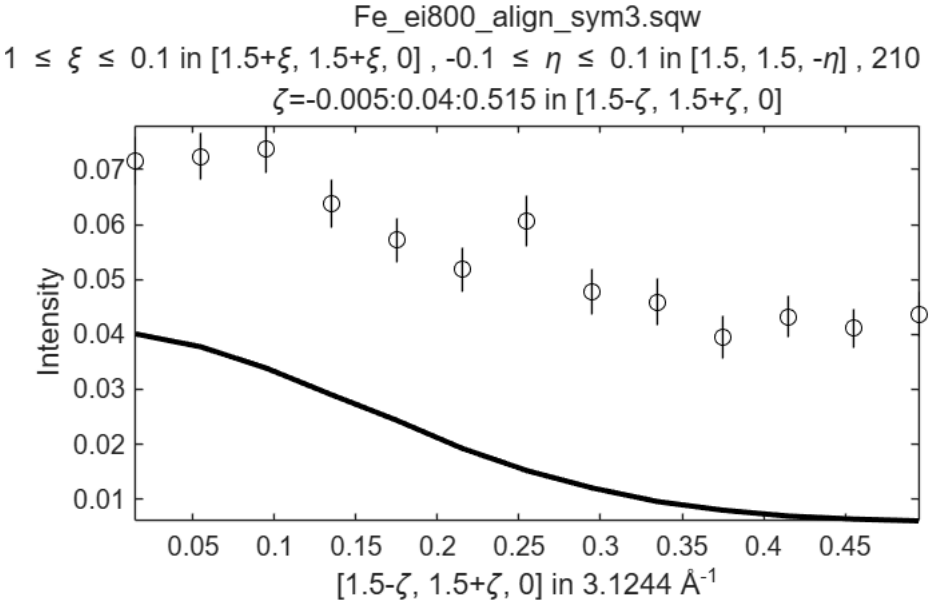
Parameter values (free parameters with error estimates):
---------------------------------------------------------

Foreground parameter values:
----------------------------
    1              1
    2              8
    3           1509   +/-   1.48e+05
    4          6.004   +/-   348.8
    5              0
    6          115.1   +/-   5786
    7             35   +/-   7208
    8              0
    9              0
   10              0


Background parameter values:
----------------------------
    1         0.0348   +/-   0.01952
    2              0
    3              0


Covariance matrix for free parameters:
--------------------------------------
    1.0000    1.0000    1.0000   -1.0000   -0.9686
    1.0000    1.0000    1.0000   -1.0000   -0.9674
    1.0000    1.0000    1.0000   -1.0000   -0.9673
   -1.0000   -1.0000   -1.0000    1.0000    0.9678
   -0.9686   -0.9674   -0.9673    0.9678    1.0000
```






```
*****************************
**** En = 225±5 going to 345
*****************************
-------------------------------------------------------------------------------
Beginning fit (max 20 iterations)---------------------------------------------------------------------------------------------
Starting point
------------------
 Total time = 0.35306s    Reduced Chi^2 = 64.252
 Free parameter values:
        149.5          11.53          78.06             35        0.05401
-------------------------------------------------------------------------------
Iteration = 1
------------------
 Total time = 2.0036s    Reduced Chi^2 = 1.6115      Levenberg-Marquardt = 0.1
 Free parameter values:
        142.2          12.2          79.88             35        0.03273
-------------------------------------------------------------------------------
Iteration = 2
------------------
 Total time = 3.8501s    Reduced Chi^2 = 1.3814      Levenberg-Marquardt = 0.01
 Free parameter values:
        156.6          14.2          84.13             35        0.03028
-------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 5.7396s    Reduced Chi^2 = 11.4107      Levenberg-Marquardt = 0.001
 Free parameter values:
        279.5          6.252          87.74             35        0.02969
 Total time = 6.1622s    Reduced Chi^2 = 1.4904      Levenberg-Marquardt = 0.01
 Free parameter values:
          197          11.72          85.38             35        0.03006
 Total time = 6.5416s    Reduced Chi^2 = 1.3678      Levenberg-Marquardt = 0.1
 Free parameter values:
        169.7          13.45          84.48             35        0.03019
-------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 8.4477s    Reduced Chi^2 = 1.3767      Levenberg-Marquardt = 0.01
 Free parameter values:
        173.4          17.19          91.79             35        0.03003
 Total time = 8.8246s    Reduced Chi^2 = 1.354      Levenberg-Marquardt = 0.1
 Free parameter values:
```

```
        171.2          14.65          86.76             35        0.03014
--------------------------------------------------------------------------------
Iteration = 5
------------------
 Total time = 10.7199s    Reduced Chi^2 = 1.3438      Levenberg-Marquardt = 0.01
 Free parameter values:
        186.8          15.91          91.03             35        0.02997
--------------------------------------------------------------------------------
Iteration = 6
------------------
 Total time = 12.7279s    Reduced Chi^2 = 1.8916      Levenberg-Marquardt = 0.001
 Free parameter values:
        197.9          27.19          111.9             35         0.0296
 Total time = 13.1016s    Reduced Chi^2 = 1.3541      Levenberg-Marquardt = 0.01
 Free parameter values:
        191.3          19.48          97.74             35        0.02985
 Total time = 13.5124s    Reduced Chi^2 = 1.3667      Levenberg-Marquardt = 0.1
 Free parameter values:
        188.4          17.04          93.14             35        0.02993
 Total time = 13.9115s    Reduced Chi^2 = 1.3479      Levenberg-Marquardt = 10
 Free parameter values:
          187          16.03          91.23             35        0.02997
 Total time = 14.2957s    Reduced Chi^2 = 1.3858      Levenberg-Marquardt = 1000
 Free parameter values:
        186.8          15.92          91.05             35        0.02997
 Total time = 14.6839s    Reduced Chi^2 = 1.3622      Levenberg-Marquardt = 100000
 Free parameter values:
        186.8          15.91          91.03             35        0.02997
 *** No improvement in chi-squared over previous iteration ***
 Total time = 14.6927s    Reduced Chi^2 = 1.3438
 Free parameter values:
        186.8          15.91          91.03             35        0.02997
--------------------------------------------------------------------------------
Fit converged:

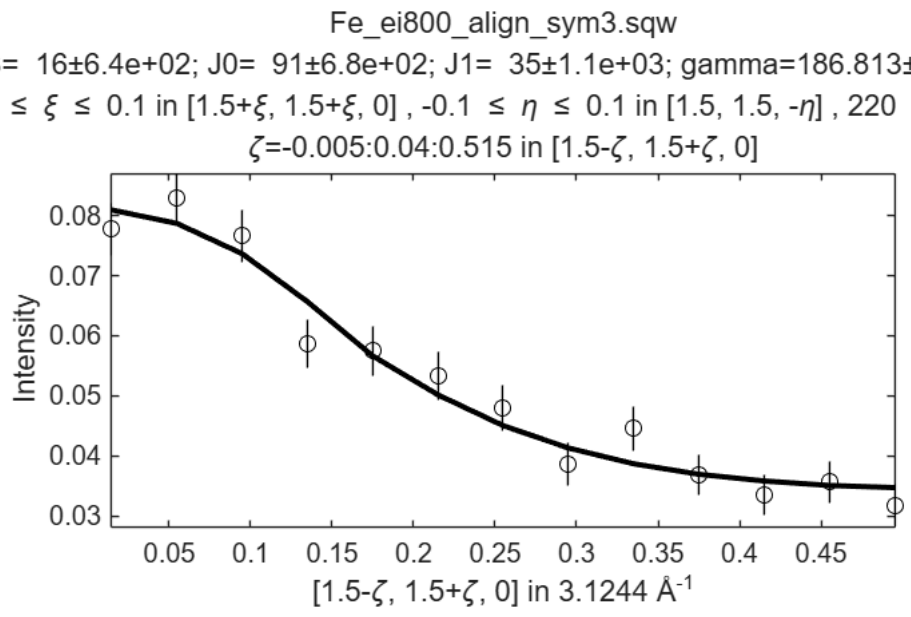
Parameter values (free parameters with error estimates):
--------------------------------------------------------

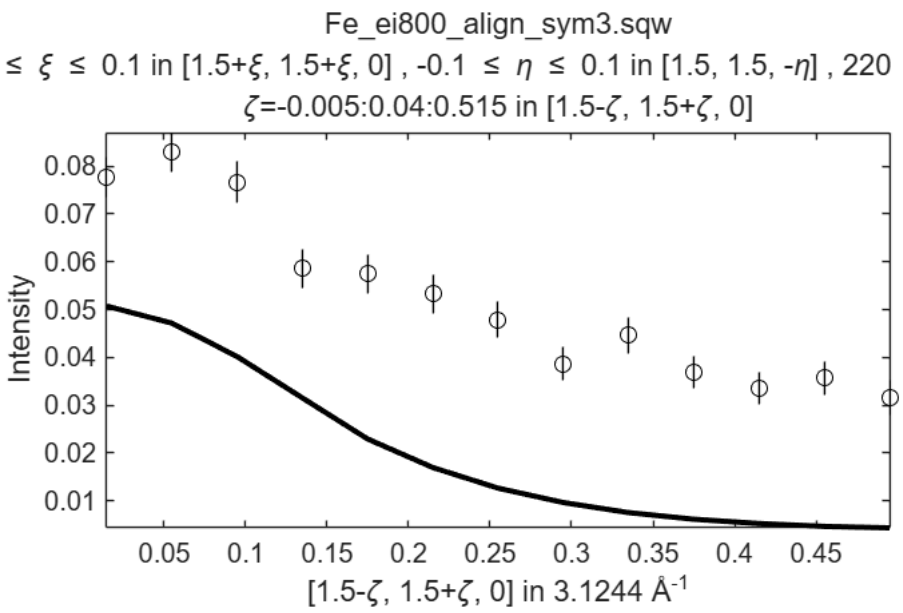
Foreground parameter values:
----------------------------
    1              1
    2              8
    3          186.8   +/-   3377
    4          15.91   +/-   643.5
    5              0
    6          91.03   +/-   683
    7             35   +/-   1082
    8              0
    9              0
   10              0


Background parameter values:
----------------------------
    1        0.02997   +/-   0.005162
    2              0
    3              0


Covariance matrix for free parameters:
--------------------------------------
    1.0000   -0.9659   -0.9073    0.9001   -0.1767
   -0.9659    1.0000    0.9852   -0.9822   -0.0660
   -0.9073    0.9852    1.0000   -0.9999   -0.2232
```

```
    0.9001   -0.9822   -0.9999    1.0000    0.2383
   -0.1767   -0.0660   -0.2232    0.2383    1.0000
```






```
*****************************
**** En = 235±5 going to 345
*****************************
-------------------------------------------------------------------------------
Beginning fit (max 20 iterations)-------------------------------------------------------------------------------------------
Starting point
------------------
 Total time = 0.38671s    Reduced Chi^2 = 94.1702
 Free parameter values:
         166.6          10.91          77.59             35        0.05073
-------------------------------------------------------------------------------
Iteration = 1
------------------
 Total time = 2.3195s    Reduced Chi^2 = 2.4755      Levenberg-Marquardt = 0.1
 Free parameter values:
           168          9.169          75.52             35        0.02677
-------------------------------------------------------------------------------
Iteration = 2
------------------
```

```
 Total time = 4.5557s    Reduced Chi^2 = 2.1255      Levenberg-Marquardt = 0.01
 Free parameter values:
        185.3           8.44          73.73             35        0.02414
------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 6.7983s    Reduced Chi^2 = 18.7173      Levenberg-Marquardt = 0.001
 Free parameter values:
        211.7          2.378           60.2             35        0.02406
 Total time = 7.2498s    Reduced Chi^2 = 2.1264      Levenberg-Marquardt = 0.01
 Free parameter values:
        195.3          6.514          69.49             35        0.02407
 Total time = 7.7328s    Reduced Chi^2 = 2.0828      Levenberg-Marquardt = 0.1
 Free parameter values:
        188.8          7.841          72.36             35        0.02409
------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 9.9006s    Reduced Chi^2 = 2.1146      Levenberg-Marquardt = 0.01
 Free parameter values:
        196.2          6.493          69.19             35        0.02402
 Total time = 10.3274s    Reduced Chi^2 = 2.1346      Levenberg-Marquardt = 0.1
 Free parameter values:
        191.4          7.419          71.34             35        0.02405
 Total time = 10.7506s    Reduced Chi^2 = 2.089      Levenberg-Marquardt = 1
 Free parameter values:
        189.7          7.712          72.03             35        0.02408
 Total time = 11.2334s    Reduced Chi^2 = 2.1049      Levenberg-Marquardt = 100
 Free parameter values:
        188.9          7.829          72.33             35        0.02409
 Total time = 11.7612s    Reduced Chi^2 = 2.0852      Levenberg-Marquardt = 10000
 Free parameter values:
        188.8           7.84          72.36             35        0.02409
 Total time = 12.2083s    Reduced Chi^2 = 2.108      Levenberg-Marquardt = 1000000
 Free parameter values:
        188.8          7.841          72.36             35        0.02409
 *** No improvement in chi-squared over previous iteration ***
 Total time = 12.216s    Reduced Chi^2 = 2.0828
 Free parameter values:
        188.8          7.841          72.36             35        0.02409
------------------------------------------------------------------------------
Fit converged:

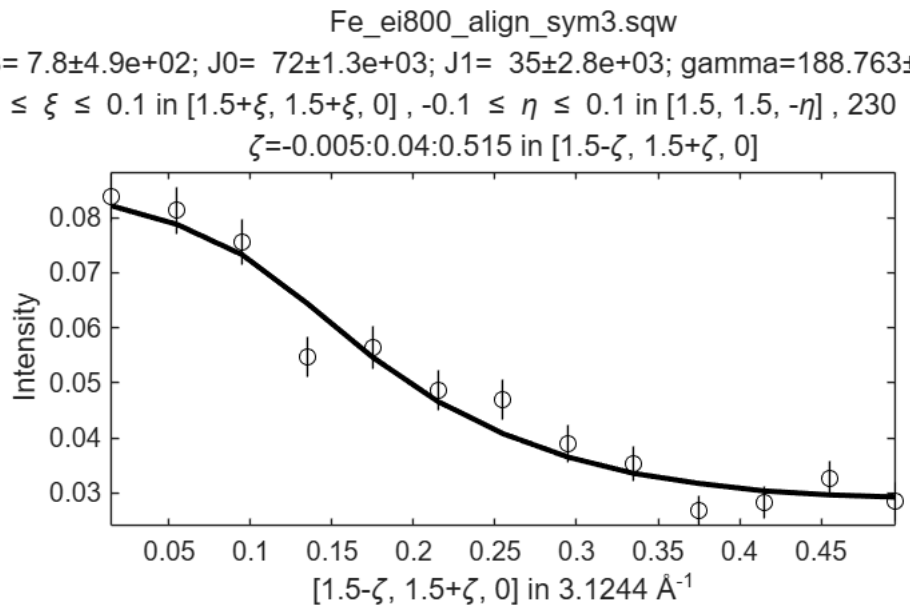
Parameter values (free parameters with error estimates):
---------------------------------------------------------

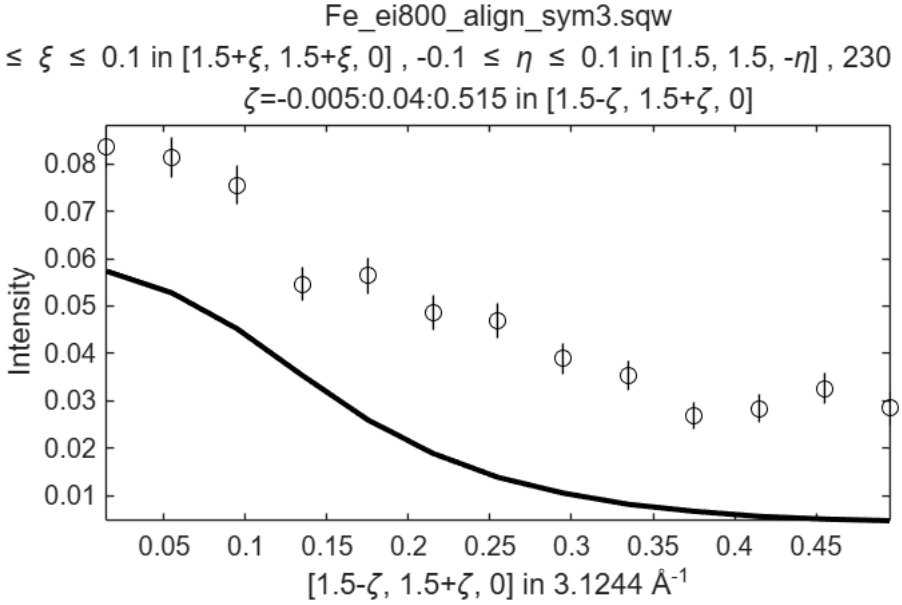
Foreground parameter values:
----------------------------
    1              1
    2              8
    3          188.8   +/-   507.9
    4          7.841   +/-   490.7
    5              0
    6          72.36   +/-   1335
    7             35   +/-   2751
    8              0
    9              0
   10              0


Background parameter values:
----------------------------
    1        0.02409   +/-   0.01289
    2              0
```

```
    3              0


Covariance matrix for free parameters:
-------------------------------------
    1.0000   -0.7303   -0.7146    0.7118    0.4821
   -0.7303    1.0000    0.9997   -0.9996   -0.9393
   -0.7146    0.9997    1.0000   -1.0000   -0.9456
    0.7118   -0.9996   -1.0000    1.0000    0.9467
    0.4821   -0.9393   -0.9456    0.9467    1.0000
```






```
*****************************
**** En = 245±5 going to 345
*****************************
--------------------------------------------------------------------------------
Beginning fit (max 20 iterations)----------------------------------------------------------------------------------------
Starting point
------------------
 Total time = 0.47929s    Reduced Chi^2 = 101.5754
 Free parameter values:
         148.4          7.753           69.9             35        0.04706
--------------------------------------------------------------------------------
```

```
Iteration = 1
------------------
 Total time = 2.7403s    Reduced Chi^2 = 1.727      Levenberg-Marquardt = 0.1
 Free parameter values:
         134.3          7.907          70.13              35        0.02439
-------------------------------------------------------------------------------
Iteration = 2
------------------
 Total time = 5.0932s    Reduced Chi^2 = 1.3583      Levenberg-Marquardt = 0.01
 Free parameter values:
         149.5          6.324          66.59              35        0.02201
-------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 7.3328s    Reduced Chi^2 = 1.5656      Levenberg-Marquardt = 0.001
 Free parameter values:
         140.3          8.678          72.19              35        0.02187
 Total time = 7.7991s    Reduced Chi^2 = 1.3172      Levenberg-Marquardt = 0.01
 Free parameter values:
           147          7.121          68.19              35         0.0219
-------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 10.1415s    Reduced Chi^2 = 1.3001      Levenberg-Marquardt = 0.001
 Free parameter values:
         146.6          6.413          66.11              35        0.02196
-------------------------------------------------------------------------------
Iteration = 5
------------------
 Total time = 12.5532s    Reduced Chi^2 = 8.6285      Levenberg-Marquardt = 0.0001
 Free parameter values:
           120          14.02          85.48              35        0.02168
 Total time = 13.0182s    Reduced Chi^2 = 1.5045      Levenberg-Marquardt = 0.001
 Free parameter values:
         137.4          8.843          72.15              35        0.02187
 Total time = 13.4878s    Reduced Chi^2 = 1.317      Levenberg-Marquardt = 0.01
 Free parameter values:
         143.6          7.194          67.97              35        0.02193
 Total time = 14.0279s    Reduced Chi^2 = 1.3035      Levenberg-Marquardt = 1
 Free parameter values:
         146.5            6.5          66.26              35        0.02196
 Total time = 14.4993s    Reduced Chi^2 = 1.2999      Levenberg-Marquardt = 100
 Free parameter values:
         146.6          6.423          66.12              35        0.02196
-------------------------------------------------------------------------------
Fit converged:

Parameter values (free parameters with error estimates):
---------------------------------------------------------

Foreground parameter values:
----------------------------
    1              1
    2              8
    3          146.6   +/-   NaN
    4          6.423   +/-   NaN
    5              0
    6          66.12   +/-   NaN
    7             35   +/-   NaN
    8              0
    9              0
   10              0
```

```
Background parameter values:
----------------------------
    1        0.02196   +/-   NaN
    2              0
    3              0

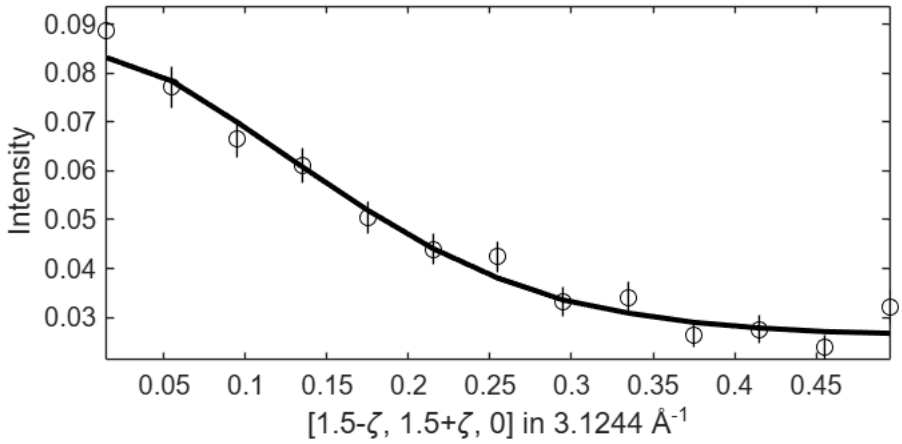

Covariance matrix for free parameters:
--------------------------------------
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
```

Fe_ei800_align_sym3.sqw

S= 6.4± NaN; J0= 66± NaN; J1= 35± NaN; gamma=146.607±NaN

1 ≤ ξ ≤ 0.1 in [1.5+ξ, 1.5+ξ, 0] , -0.1 ≤ η ≤ 0.1 in [1.5, 1.5, -η] , 240 ≤ E ≤

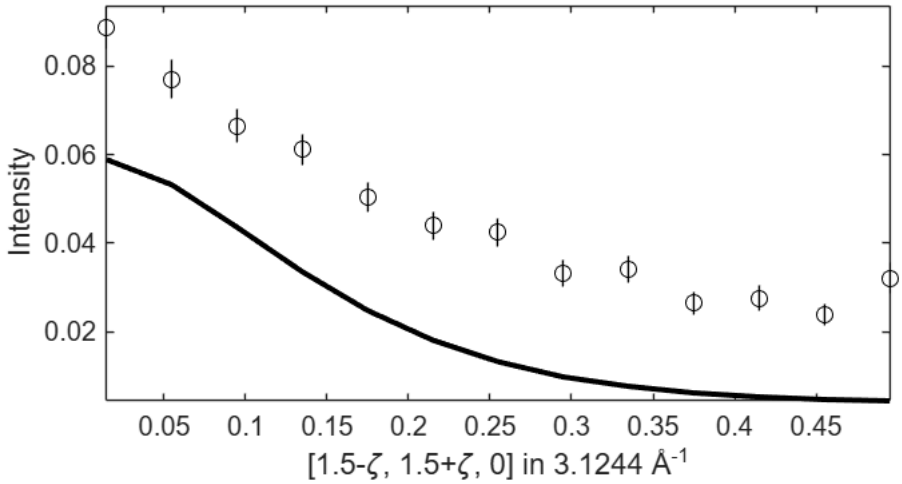
ζ=-0.005:0.04:0.515 in [1.5-ζ, 1.5+ζ, 0]



Fe_ei800_align_sym3.sqw

1 ≤ ξ ≤ 0.1 in [1.5+ξ, 1.5+ξ, 0] , -0.1 ≤ η ≤ 0.1 in [1.5, 1.5, -η] , 240 ≤ E ≤

ζ=-0.005:0.04:0.515 in [1.5-ζ, 1.5+ζ, 0]




```
*****************************
**** En = 255±5 going to 345
*****************************
--------------------------------------------------------------------------------
Beginning fit (max 20 iterations)----------------------------------------------------------------------------------------
Starting point
------------------
```

```
 Total time = 0.43953s    Reduced Chi^2 = 147.7154
 Free parameter values:
         1815           11.5          150.5             35        0.04348
-------------------------------------------------------------------------------
Iteration = 1
------------------
 Total time = 2.8058s    Reduced Chi^2 = 2.8354      Levenberg-Marquardt = 0.1
 Free parameter values:
         1519          12.73          164.8             35        0.02115
-------------------------------------------------------------------------------
Iteration = 2
------------------
 Total time = 5.142s    Reduced Chi^2 = 2.7057      Levenberg-Marquardt = 0.01
 Free parameter values:
         2688          15.37          199.4             35        0.01587
-------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 7.5692s    Reduced Chi^2 = 6.0218      Levenberg-Marquardt = 0.001
 Free parameter values:
         7001          27.05          338.9             35         0.0146
 Total time = 8.0298s    Reduced Chi^2 = 2.0571      Levenberg-Marquardt = 0.01
 Free parameter values:
         4323          18.93            241             35        0.01496
-------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 10.3978s    Reduced Chi^2 = 4.6453      Levenberg-Marquardt = 0.001
 Free parameter values:
     1.096e+04           33.5          414.6             35        0.01447
 Total time = 10.8782s    Reduced Chi^2 = 1.8116      Levenberg-Marquardt = 0.01
 Free parameter values:
         6652          23.49          294.2             35        0.01465
-------------------------------------------------------------------------------
Iteration = 5
------------------
 Total time = 13.2614s    Reduced Chi^2 = 4.0126      Levenberg-Marquardt = 0.001
 Free parameter values:
     1.637e+04          40.44          498.8             35         0.0144
 Total time = 13.7692s    Reduced Chi^2 = 1.737      Levenberg-Marquardt = 0.01
 Free parameter values:
         9977          28.74          357.6             35        0.01453
-------------------------------------------------------------------------------
Iteration = 6
------------------
 Total time = 15.8095s    Reduced Chi^2 = 3.9016      Levenberg-Marquardt = 0.001
 Free parameter values:
     2.441e+04          48.94          604.6             35        0.01434
 Total time = 16.2429s    Reduced Chi^2 = 1.7507      Levenberg-Marquardt = 0.01
 Free parameter values:
     1.488e+04          34.96          434.4             35        0.01445
 Total time = 16.6957s    Reduced Chi^2 = 1.6393      Levenberg-Marquardt = 0.1
 Free parameter values:
     1.168e+04          30.76          380.9             35        0.01462
-------------------------------------------------------------------------------
Iteration = 7
------------------
 Total time = 19.1746s    Reduced Chi^2 = 1.6878      Levenberg-Marquardt = 0.01
 Free parameter values:
      1.74e+04          37.74          468.1             35        0.01444
 Total time = 19.696s    Reduced Chi^2 = 1.6089      Levenberg-Marquardt = 0.1
 Free parameter values:
     1.357e+04          32.93          408.2             35        0.01457
-------------------------------------------------------------------------------
```

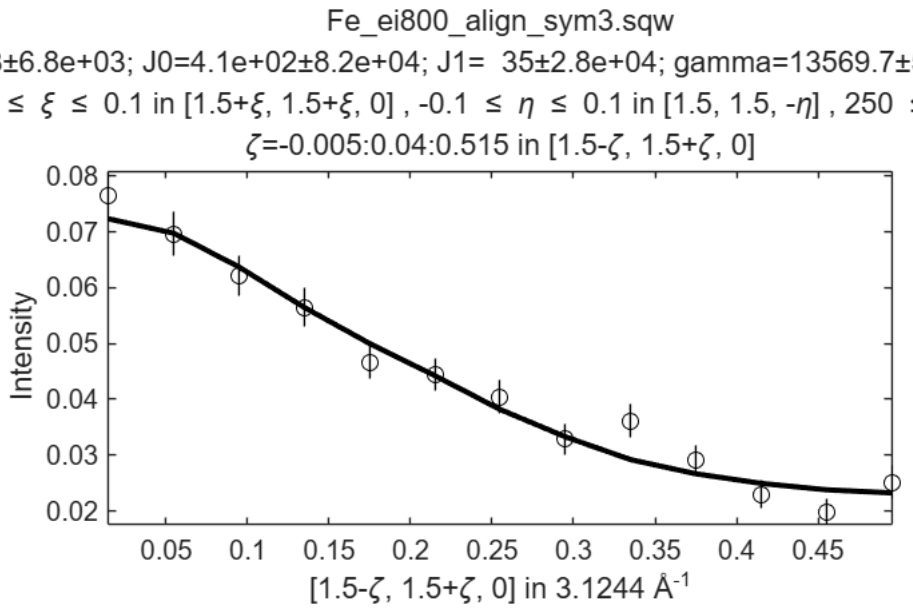
```
Iteration = 8
------------------
 Total time = 21.9865s    Reduced Chi^2 = 1.6987      Levenberg-Marquardt = 0.01
 Free parameter values:
     1.988e+04          40.35          499.2             35        0.01437
 Total time = 22.4404s    Reduced Chi^2 = 1.6158      Levenberg-Marquardt = 0.1
 Free parameter values:
     1.562e+04          35.29          436.7             35        0.01447
 Total time = 22.9131s    Reduced Chi^2 = 1.6149      Levenberg-Marquardt = 1
 Free parameter values:
     1.424e+04           33.7          417.1             35        0.01454
 Total time = 23.3791s    Reduced Chi^2 = 1.6125      Levenberg-Marquardt = 100
 Free parameter values:
     1.364e+04          33.01          409.1             35        0.01457
 Total time = 23.868s     Reduced Chi^2 = 1.6163      Levenberg-Marquardt = 10000
 Free parameter values:
     1.358e+04          32.94          408.3             35        0.01457
 Total time = 24.3513s    Reduced Chi^2 = 1.6255      Levenberg-Marquardt = 1000000
 Free parameter values:
     1.357e+04          32.93          408.2             35        0.01457
 *** No improvement in chi-squared over previous iteration ***
 Total time = 24.3588s    Reduced Chi^2 = 1.6089
 Free parameter values:
     1.357e+04          32.93          408.2             35        0.01457
--------------------------------------------------------------------------------
Fit converged:

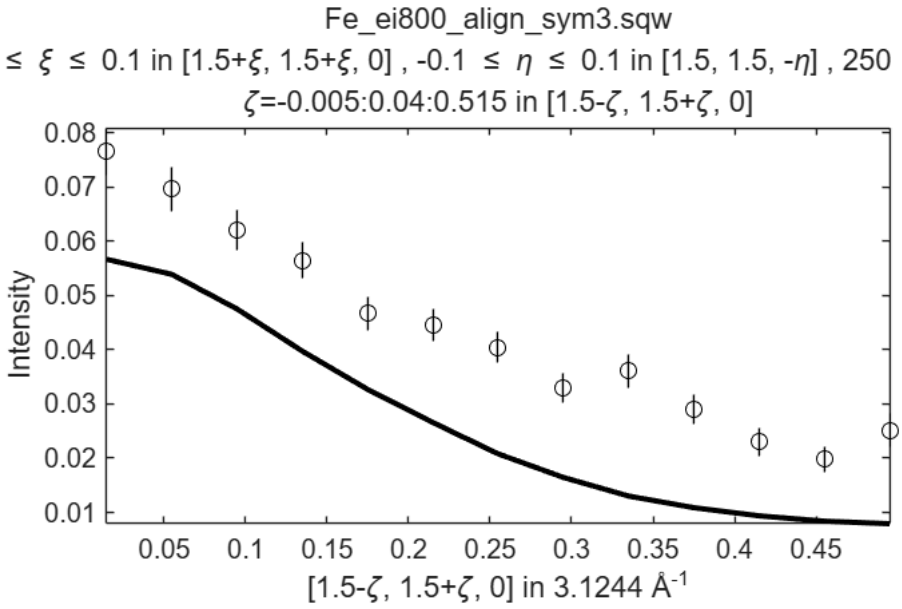
Parameter values (free parameters with error estimates):
---------------------------------------------------------

Foreground parameter values:
----------------------------
    1              1
    2              8
    3      1.357e+04   +/-   5.442e+06
    4          32.93   +/-   6788
    5              0
    6          408.2   +/-   8.232e+04
    7             35   +/-   2.828e+04
    8              0
    9              0
   10              0


Background parameter values:
----------------------------
    1        0.01457   +/-   0.01153
    2              0
    3              0


Covariance matrix for free parameters:
--------------------------------------
    1.0000    1.0000    1.0000   -1.0000   -0.9448
    1.0000    1.0000    1.0000   -1.0000   -0.9444
    1.0000    1.0000    1.0000   -1.0000   -0.9445
   -1.0000   -1.0000   -1.0000    1.0000    0.9445
   -0.9448   -0.9444   -0.9445    0.9445    1.0000
```






```
*****************************
**** En = 265±5 going to 345
*****************************
-------------------------------------------------------------------------------
Beginning fit (max 20 iterations)-------------------------------------------------------------------------------------------
Starting point
------------------
 Total time = 0.52337s    Reduced Chi^2 = 183.3203
 Free parameter values:
          7205          18.83          267.7             35        0.03986
-------------------------------------------------------------------------------
Iteration = 1
------------------
 Total time = 3.1312s    Reduced Chi^2 = 3.7963      Levenberg-Marquardt = 0.1
 Free parameter values:
          4617          18.41          287.9             35        0.02061
-------------------------------------------------------------------------------
Iteration = 2
------------------
 Total time = 5.7638s    Reduced Chi^2 = 4.2623      Levenberg-Marquardt = 0.01
 Free parameter values:
```

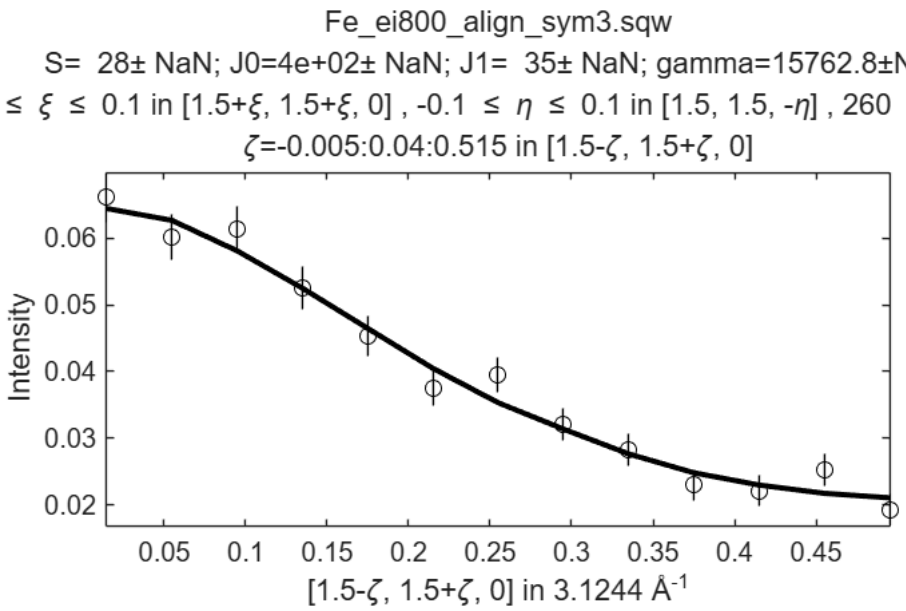
```
          8074          16.41          288.3             35        0.01366
 Total time = 6.3294s    Reduced Chi^2 = 1.4855      Levenberg-Marquardt = 0.1
 Free parameter values:
          6082          19.01          280.7             35        0.01562
-------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 8.9804s    Reduced Chi^2 = 1.3071      Levenberg-Marquardt = 0.01
 Free parameter values:
          8077          19.67          289.6             35        0.01223
-------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 11.406s    Reduced Chi^2 = 1.1896      Levenberg-Marquardt = 0.001
 Free parameter values:
     1.064e+04          23.16            329             35        0.01144
-------------------------------------------------------------------------------
Iteration = 5
------------------
 Total time = 13.88s    Reduced Chi^2 = 1.7114      Levenberg-Marquardt = 0.0001
 Free parameter values:
     2.104e+04          34.64          486.5             35        0.01137
 Total time = 14.4289s    Reduced Chi^2 = 1.1673      Levenberg-Marquardt = 0.001
 Free parameter values:
     1.406e+04          26.81          378.3             35        0.01141
-------------------------------------------------------------------------------
Iteration = 6
------------------
 Total time = 17.0688s    Reduced Chi^2 = 1.8941      Levenberg-Marquardt = 0.0001
 Free parameter values:
     2.958e+04          41.77          584.8             35        0.01145
 Total time = 17.6003s    Reduced Chi^2 = 1.1702      Levenberg-Marquardt = 0.001
 Free parameter values:
     1.909e+04          31.52          443.3             35        0.01148
 Total time = 18.1454s    Reduced Chi^2 = 1.1661      Levenberg-Marquardt = 0.01
 Free parameter values:
     1.576e+04          28.29          398.5             35        0.01149
-------------------------------------------------------------------------------
Fit converged:

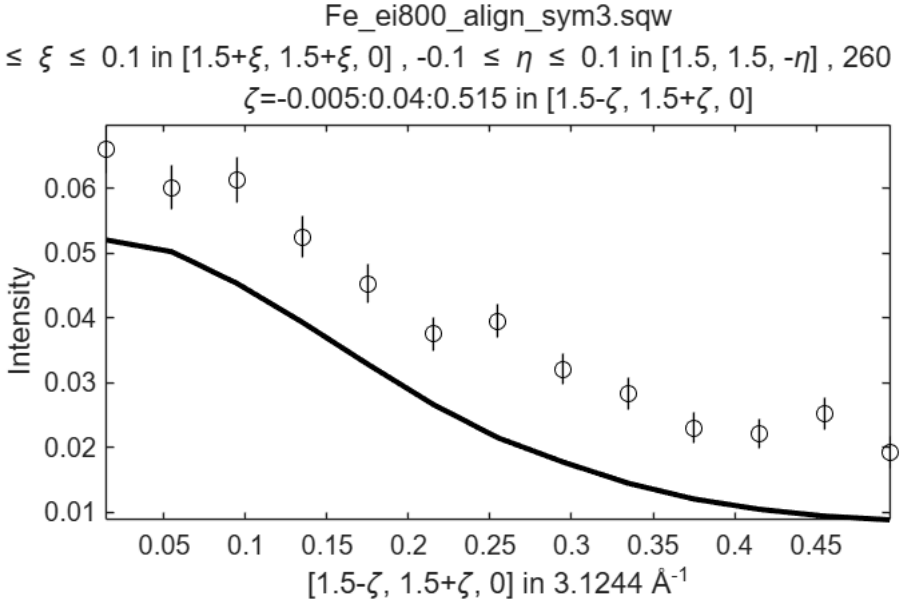
Parameter values (free parameters with error estimates):
---------------------------------------------------------

Foreground parameter values:
----------------------------
    1              1
    2              8
    3      1.576e+04   +/-   NaN
    4          28.29   +/-   NaN
    5              0
    6          398.5   +/-   NaN
    7             35   +/-   NaN
    8              0
    9              0
   10              0


Background parameter values:
----------------------------
    1        0.01149   +/-   NaN
    2              0
    3              0
```

```
Covariance matrix for free parameters:
-------------------------------------
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
```






```
*****************************
**** En = 275±5 going to 345
*****************************
--------------------------------------------------------------------------------
Beginning fit (max 20 iterations)----------------------------------------------------------------------------------------------------
Starting point
------------------
 Total time = 0.55598s    Reduced Chi^2 = 169.2581
 Free parameter values:
         3237          11.64          184.4             35        0.03854
--------------------------------------------------------------------------------
Iteration = 1
------------------
 Total time = 3.4347s    Reduced Chi^2 = 4.2009      Levenberg-Marquardt = 0.1
```

```
 Free parameter values:
          1881          11.12            194             35        0.02109
--------------------------------------------------------------------------------
Iteration = 2
------------------
 Total time = 6.2284s    Reduced Chi^2 = 5.8561      Levenberg-Marquardt = 0.01
 Free parameter values:
          3289          8.658          183.3             35        0.01493
 Total time = 6.7771s    Reduced Chi^2 = 1.5406      Levenberg-Marquardt = 0.1
 Free parameter values:
          2498          11.18          185.2             35        0.01667
--------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 9.6652s    Reduced Chi^2 = 1.3198      Levenberg-Marquardt = 0.01
 Free parameter values:
          3104          10.74          179.6             35        0.01332
--------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 12.3809s    Reduced Chi^2 = 1.1921      Levenberg-Marquardt = 0.001
 Free parameter values:
          2748          10.58          168.7             35        0.01262
--------------------------------------------------------------------------------
Iteration = 5
------------------
 Total time = 15.1546s    Reduced Chi^2 = 1.1814      Levenberg-Marquardt = 0.0001
 Free parameter values:
          2338          9.629          155.7             35        0.01267
--------------------------------------------------------------------------------
Iteration = 6
------------------
 Total time = 17.9808s    Reduced Chi^2 = 1.3201      Levenberg-Marquardt = 1e-05
 Free parameter values:
          1373          7.329          122.8             35        0.01281
 Total time = 18.5315s    Reduced Chi^2 = 1.1798      Levenberg-Marquardt = 0.0001
 Free parameter values:
          2032           8.88            145             35        0.01274
--------------------------------------------------------------------------------
Iteration = 7
------------------
 Total time = 21.2647s    Reduced Chi^2 = 1.1805      Levenberg-Marquardt = 1e-05
 Free parameter values:
          1596          7.776          129.1             35        0.01281
 Total time = 21.8404s    Reduced Chi^2 = 1.1751      Levenberg-Marquardt = 0.0001
 Free parameter values:
          1891          8.528          139.8             35        0.01277
--------------------------------------------------------------------------------
Iteration = 8
------------------
 Total time = 24.5463s    Reduced Chi^2 = 1.1971      Levenberg-Marquardt = 1e-05
 Free parameter values:
          1554          7.606          126.9             35        0.01282
 Total time = 25.1907s    Reduced Chi^2 = 1.1843      Levenberg-Marquardt = 0.0001
 Free parameter values:
          1785          8.218          135.6             35        0.01279
 Total time = 25.7289s    Reduced Chi^2 = 1.1919      Levenberg-Marquardt = 0.001
 Free parameter values:
          1861          8.422          138.5             35        0.01277
 Total time = 26.2616s    Reduced Chi^2 = 1.196      Levenberg-Marquardt = 0.1
 Free parameter values:
          1892          8.508          139.7             35        0.01277
 Total time = 26.7917s    Reduced Chi^2 = 1.1808      Levenberg-Marquardt = 10
 Free parameter values:
```

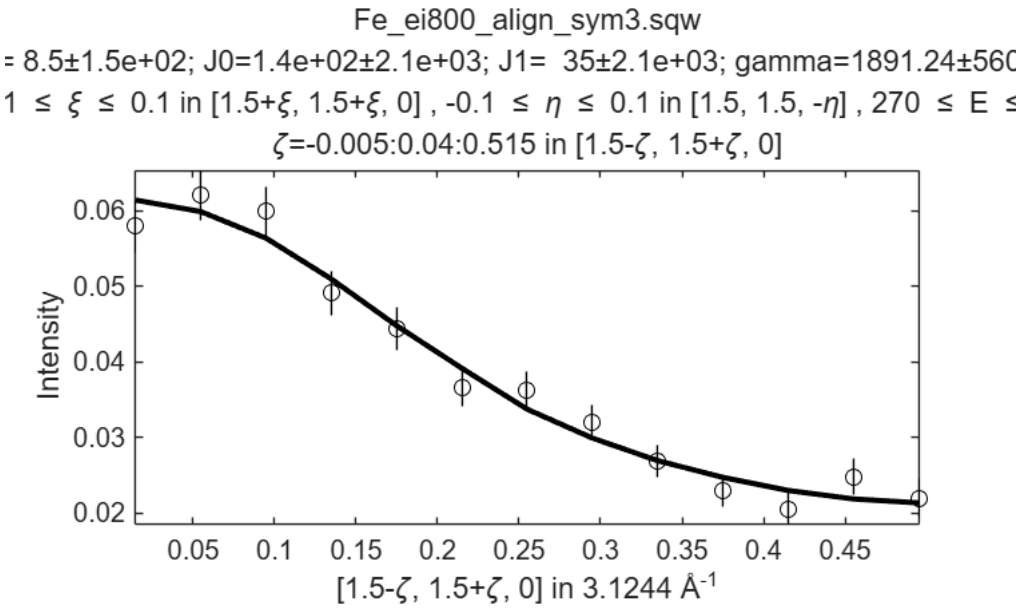
```
         1892          8.525          139.8             35        0.01277
 Total time = 27.3827s    Reduced Chi^2 = 1.1942      Levenberg-Marquardt = 1000
 Free parameter values:
         1891          8.527          139.8             35        0.01277
 *** No improvement in chi-squared over previous iteration ***
 Total time = 27.3916s    Reduced Chi^2 = 1.1751
 Free parameter values:
         1891          8.528          139.8             35        0.01277
--------------------------------------------------------------------------------
Fit converged:

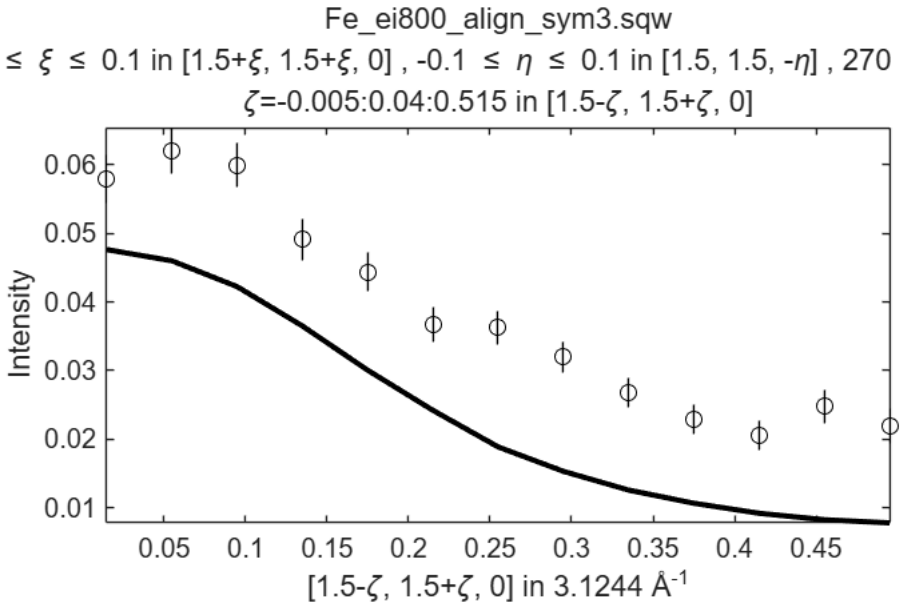
Parameter values (free parameters with error estimates):
-------------------------------------------------------

Foreground parameter values:
----------------------------
    1              1
    2              8
    3           1891   +/-   5.601e+04
    4          8.528   +/-   148.5
    5              0
    6          139.8   +/-   2105
    7             35   +/-   2148
    8              0
    9              0
   10              0


Background parameter values:
----------------------------
    1        0.01277   +/-   0.009767
    2              0
    3              0


Covariance matrix for free parameters:
--------------------------------------
    1.0000    1.0000    1.0000   -1.0000   -0.9470
    1.0000    1.0000    1.0000   -1.0000   -0.9455
    1.0000    1.0000    1.0000   -1.0000   -0.9447
   -1.0000   -1.0000   -1.0000    1.0000    0.9447
   -0.9470   -0.9455   -0.9447    0.9447    1.0000
```






```
*****************************
**** En = 285±5 going to 345
*****************************
-------------------------------------------------------------------------------
Beginning fit (max 20 iterations)---------------------------------------------------------------------------------------------
Starting point
------------------
 Total time = 0.5538s    Reduced Chi^2 = 224.7143
 Free parameter values:
        637.9          3.347          68.03             35        0.03494
-------------------------------------------------------------------------------
Iteration = 1
------------------
 Total time = 3.5961s    Reduced Chi^2 = 3.8579      Levenberg-Marquardt = 0.1
 Free parameter values:
        482.7          2.371          71.54             35        0.01815
-------------------------------------------------------------------------------
Iteration = 2
------------------
 Total time = 6.6325s    Reduced Chi^2 = 7.9501      Levenberg-Marquardt = 0.01
 Free parameter values:
        425.1          1.989          47.82             35        0.01041
 Total time = 7.2945s    Reduced Chi^2 = 2.3785      Levenberg-Marquardt = 0.1
 Free parameter values:
        487.3           2.41          61.03             35        0.01291
-------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 10.4752s    Reduced Chi^2 = 3.7336      Levenberg-Marquardt = 0.01
 Free parameter values:
        239.5           1.76          46.19             35        0.01044
 Total time = 11.1201s    Reduced Chi^2 = 1.7038      Levenberg-Marquardt = 0.1
 Free parameter values:
        380.2           2.25          55.94             35        0.01117
-------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 14.1782s    Reduced Chi^2 = 1.5191      Levenberg-Marquardt = 0.01
 Free parameter values:
          258          1.613          45.66             35        0.01063
-------------------------------------------------------------------------------
Iteration = 5
```

```
------------------
 Total time = 17.3748s    Reduced Chi^2 = 5.2286      Levenberg-Marquardt = 0.001
 Free parameter values:
        58.01         0.4297           32.8             35        0.01476
 Total time = 18.0004s    Reduced Chi^2 = 1.4563      Levenberg-Marquardt = 0.01
 Free parameter values:
        188.5           1.19          41.34             35         0.0123
-------------------------------------------------------------------------------
Iteration = 6
------------------
 Total time = 21.1602s    Reduced Chi^2 = 1.9223      Levenberg-Marquardt = 0.001
 Free parameter values:
        90.87         0.5739          34.38             35        0.01534
 Total time = 21.7995s    Reduced Chi^2 = 1.3673      Levenberg-Marquardt = 0.01
 Free parameter values:
        150.9          0.948          38.51             35        0.01358
-------------------------------------------------------------------------------
Iteration = 7
------------------
 Total time = 25.0012s    Reduced Chi^2 = 1.3763      Levenberg-Marquardt = 0.001
 Free parameter values:
        103.5         0.6496          35.43             35        0.01566
 Total time = 25.6362s    Reduced Chi^2 = 1.3365      Levenberg-Marquardt = 0.01
 Free parameter values:
        129.6         0.8145          37.15             35        0.01459
-------------------------------------------------------------------------------
Iteration = 8
------------------
 Total time = 28.7137s    Reduced Chi^2 = 1.3165      Levenberg-Marquardt = 0.001
 Free parameter values:
          110         0.6902          35.92             35        0.01565
-------------------------------------------------------------------------------
Iteration = 9
------------------
 Total time = 31.8129s    Reduced Chi^2 = 1.3176      Levenberg-Marquardt = 0.0001
 Free parameter values:
        108.9         0.6859          35.87             35        0.01575
 Total time = 32.4221s    Reduced Chi^2 = 1.3178      Levenberg-Marquardt = 0.001
 Free parameter values:
          109         0.6865          35.87             35        0.01575
 Total time = 33.0423s    Reduced Chi^2 = 1.3256      Levenberg-Marquardt = 0.01
 Free parameter values:
        109.3         0.6887           35.9             35        0.01573
 Total time = 33.6793s    Reduced Chi^2 = 1.3279      Levenberg-Marquardt = 1
 Free parameter values:
        109.7          0.691          35.91             35        0.01569
 Total time = 34.3035s    Reduced Chi^2 = 1.3245      Levenberg-Marquardt = 100
 Free parameter values:
        109.9         0.6904          35.92             35        0.01566
 Total time = 35.0055s    Reduced Chi^2 = 1.3167      Levenberg-Marquardt = 10000
 Free parameter values:
        109.9         0.6902          35.92             35        0.01565
 *** No improvement in chi-squared over previous iteration ***
 Total time = 35.0144s    Reduced Chi^2 = 1.3165
 Free parameter values:
          110         0.6902          35.92             35        0.01565
-------------------------------------------------------------------------------
Fit converged:

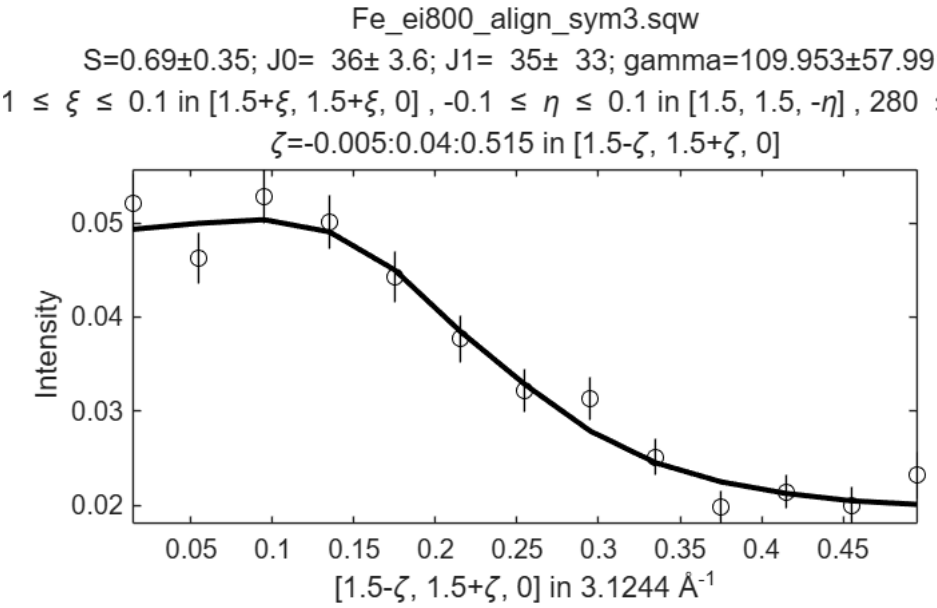
Parameter values (free parameters with error estimates):
---------------------------------------------------------

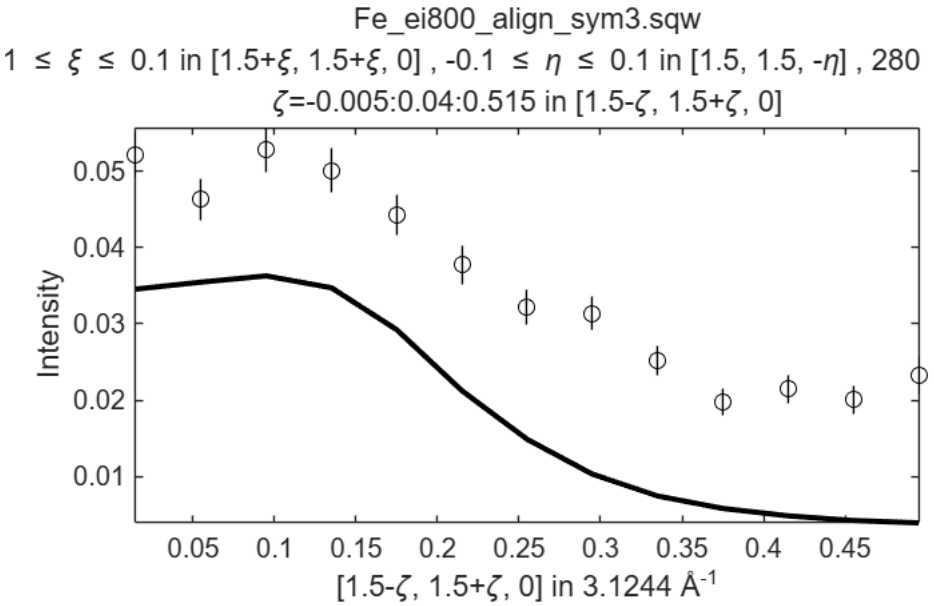
Foreground parameter values:
----------------------------
```

```
    1              1
    2              8
    3            110   +/-   57.99
    4         0.6902   +/-   0.3532
    5              0
    6          35.92   +/-   3.564
    7             35   +/-   32.57
    8              0
    9              0
   10              0


Background parameter values:
----------------------------
    1        0.01565   +/-   0.00355
    2              0
    3              0


Covariance matrix for free parameters:
--------------------------------------
    1.0000    0.9907    0.9504   -0.9949   -0.8903
    0.9907    1.0000    0.9360   -0.9840   -0.9290
    0.9504    0.9360    1.0000   -0.9728   -0.7662
   -0.9949   -0.9840   -0.9728    1.0000    0.8527
   -0.8903   -0.9290   -0.7662    0.8527    1.0000
```






```
*****************************
**** En = 295±5 going to 345
*****************************
-------------------------------------------------------------------------------
Beginning fit (max 20 iterations)---------------------------------------------------------------------------------------------
Starting point
------------------
 Total time = 0.69986s    Reduced Chi^2 = 436.5438
 Free parameter values:
          2168          7.642          107.5             35        0.03337
-------------------------------------------------------------------------------
Iteration = 1
------------------
 Total time = 4.0296s    Reduced Chi^2 = 7.212      Levenberg-Marquardt = 0.1
 Free parameter values:
          2295          5.783          127.9             35        0.01369
-------------------------------------------------------------------------------
Iteration = 2
------------------
 Total time = 7.5046s    Reduced Chi^2 = 2.8446      Levenberg-Marquardt = 0.01
 Free parameter values:
          2761          7.787          121.7             35       0.002452
-------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 10.881s    Reduced Chi^2 = 2.4387      Levenberg-Marquardt = 0.001
 Free parameter values:
          3616          10.09          139.6             35      -0.001467
-------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 14.2365s    Reduced Chi^2 = 3.6549      Levenberg-Marquardt = 0.0001
 Free parameter values:
          7373          15.91          209.3             35      -0.002308
 Total time = 14.9189s    Reduced Chi^2 = 2.3595      Levenberg-Marquardt = 0.001
 Free parameter values:
          4819          12.01          160.8             35      -0.001853
-------------------------------------------------------------------------------
Iteration = 5
------------------
 Total time = 18.2933s    Reduced Chi^2 = 3.6432      Levenberg-Marquardt = 0.0001
 Free parameter values:
```

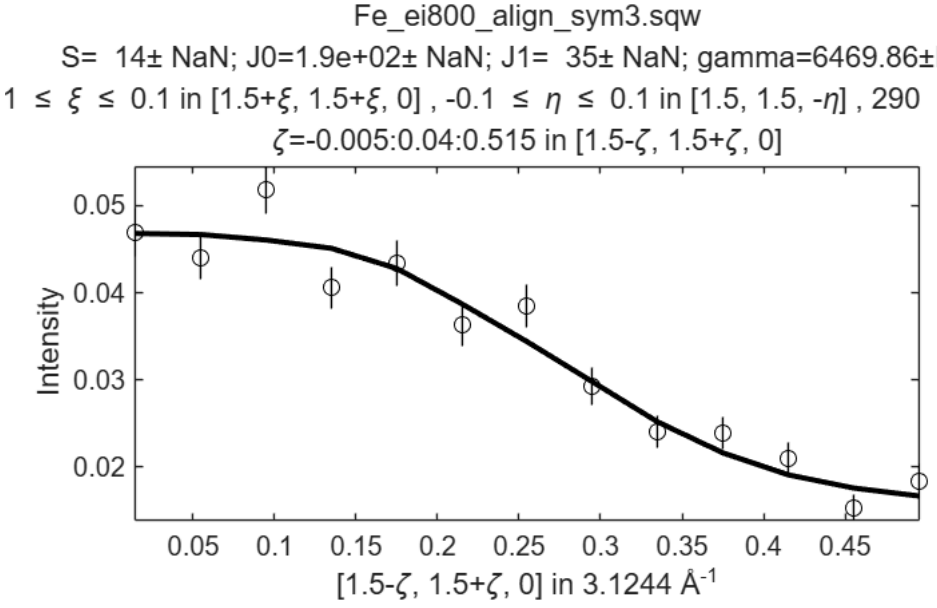
```
          9860          18.64            242              35      -0.002402
 Total time = 19.0027s    Reduced Chi^2 = 2.3583      Levenberg-Marquardt = 0.001
 Free parameter values:
          6470          14.11          186.1              35      -0.002061
------------------------------------------------------------------------------
Fit converged:

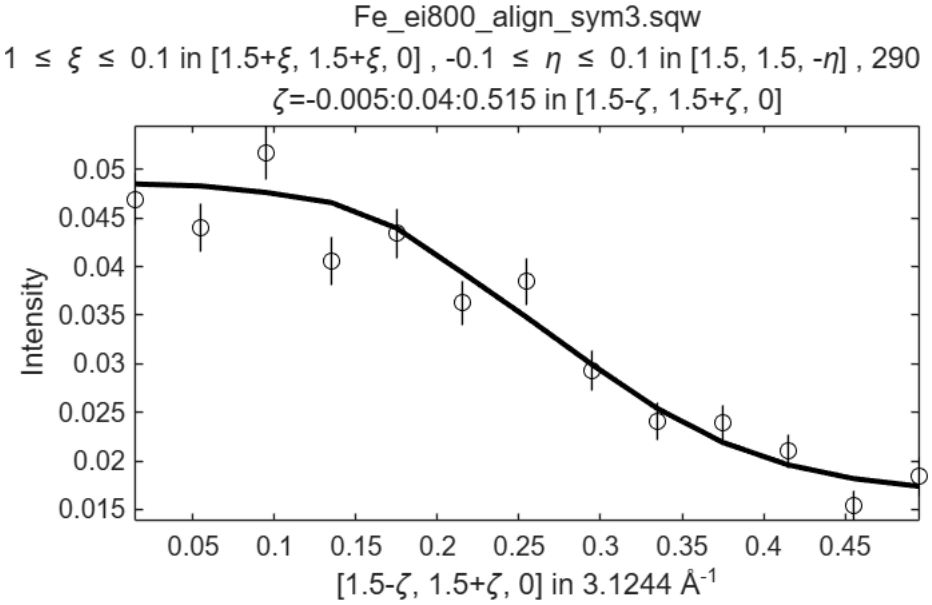
Parameter values (free parameters with error estimates):
---------------------------------------------------------

Foreground parameter values:
----------------------------
    1              1
    2              8
    3           6470   +/-   NaN
    4          14.11   +/-   NaN
    5              0
    6          186.1   +/-   NaN
    7             35   +/-   NaN
    8              0
    9              0
   10              0


Background parameter values:
----------------------------
    1      -0.002061   +/-   NaN
    2              0
    3              0


Covariance matrix for free parameters:
--------------------------------------
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
```





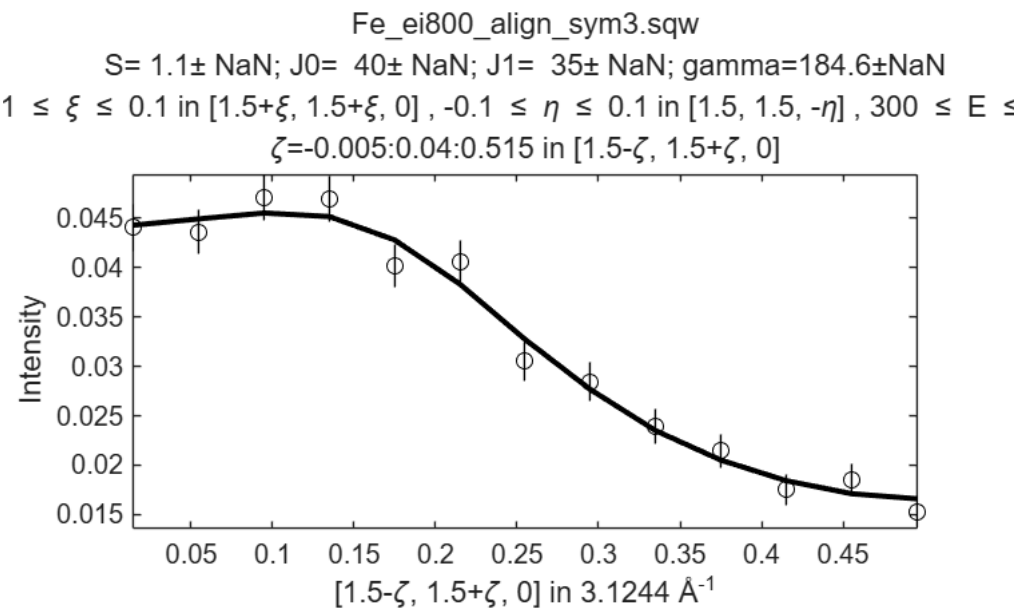

```
*****************************
**** En = 305±5 going to 345
*****************************
-------------------------------------------------------------------------------
Beginning fit (max 20 iterations)---------------------------------------------------------------------------------------------
Starting point
------------------
 Total time = 0.72205s    Reduced Chi^2 = 264.5535
 Free parameter values:
         184.9          1.125          40.44             35        0.03324
-------------------------------------------------------------------------------
Iteration = 1
------------------
 Total time = 4.6503s    Reduced Chi^2 = 2.7111      Levenberg-Marquardt = 0.1
 Free parameter values:
         171.9         0.8807          42.14             35        0.01476
-------------------------------------------------------------------------------
Iteration = 2
------------------
 Total time = 8.8167s    Reduced Chi^2 = 1.1201      Levenberg-Marquardt = 0.01
 Free parameter values:
         167.5         0.9929          38.92             35       0.009915
-------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 12.8002s    Reduced Chi^2 = 0.94247      Levenberg-Marquardt = 0.001
 Free parameter values:
         180.7          1.096          40.17             35       0.008884
-------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 16.8596s    Reduced Chi^2 = 0.92883      Levenberg-Marquardt = 0.0001
 Free parameter values:
         183.2          1.114          40.34             35       0.008798
-------------------------------------------------------------------------------
Iteration = 5
------------------
 Total time = 20.8645s    Reduced Chi^2 = 0.9283      Levenberg-Marquardt = 1e-05
 Free parameter values:
         184.6          1.122          40.42             35       0.008755
-------------------------------------------------------------------------------
Fit converged:
```

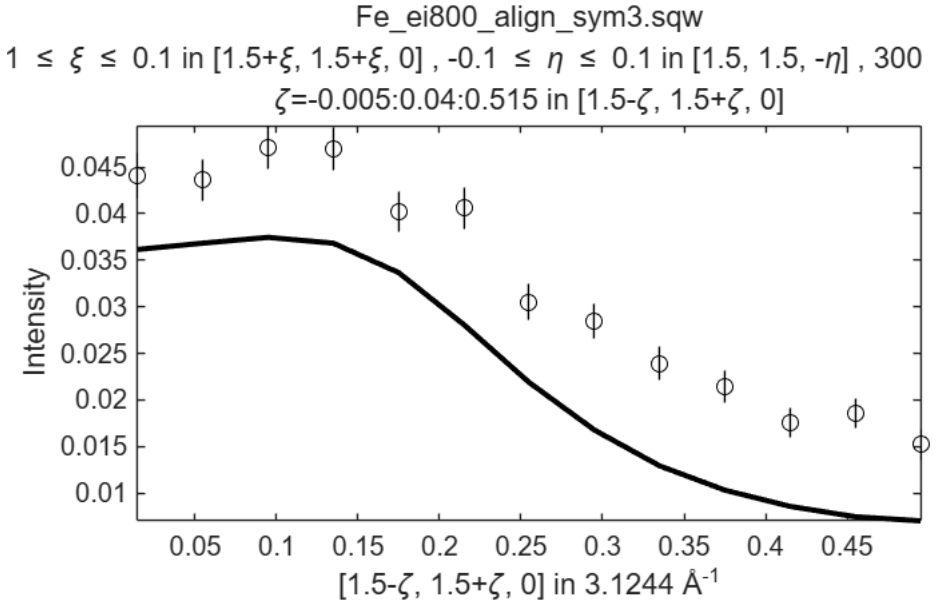
```
Parameter values (free parameters with error estimates):
-------------------------------------------------------

Foreground parameter values:
----------------------------
    1              1
    2              8
    3          184.6   +/-   NaN
    4          1.122   +/-   NaN
    5              0
    6          40.42   +/-   NaN
    7             35   +/-   NaN
    8              0
    9              0
   10              0


Background parameter values:
----------------------------
    1       0.008755   +/-   NaN
    2              0
    3              0


Covariance matrix for free parameters:
--------------------------------------
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
   NaN   NaN   NaN   NaN   NaN
```






```
*****************************
**** En = 315±5 going to 345
*****************************
---------------------------------------------------------------------------
Beginning fit (max 20 iterations)-----------------------------------------------------------------------------------------
Starting point
------------------
 Total time = 0.85076s    Reduced Chi^2 = 292.2664
 Free parameter values:
           275          1.501          43.95             35        0.03197
---------------------------------------------------------------------------
Iteration = 1
------------------
 Total time = 5.1167s    Reduced Chi^2 = 3.118      Levenberg-Marquardt = 0.1
 Free parameter values:
         271.6          1.127          47.35             35        0.01527
---------------------------------------------------------------------------
Iteration = 2
------------------
 Total time = 9.2806s    Reduced Chi^2 = 1.6897      Levenberg-Marquardt = 0.01
 Free parameter values:
         240.4           1.28          41.07             35       0.008768
---------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 13.5693s    Reduced Chi^2 = 0.90899      Levenberg-Marquardt = 0.001
 Free parameter values:
         264.3          1.436          43.36             35       0.007268
---------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 18.0056s    Reduced Chi^2 = 0.88338      Levenberg-Marquardt = 0.0001
 Free parameter values:
         273.9          1.495          43.88             35       0.006988
---------------------------------------------------------------------------
Iteration = 5
------------------
 Total time = 22.286s    Reduced Chi^2 = 0.91878      Levenberg-Marquardt = 1e-05
 Free parameter values:
         278.2           1.52          44.08             35       0.006836
 Total time = 23.1702s    Reduced Chi^2 = 0.90938      Levenberg-Marquardt = 0.0001
 Free parameter values:
```

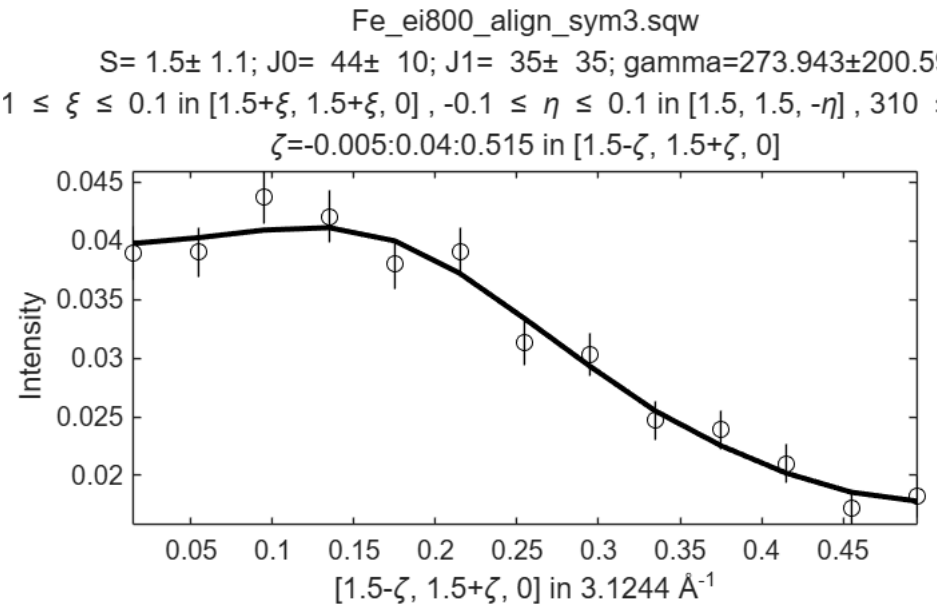
```
        277.8          1.518          44.06             35       0.006847
 Total time = 24.076s    Reduced Chi^2 = 0.90101      Levenberg-Marquardt = 0.001
 Free parameter values:
        276.2          1.509          43.98             35       0.006891
 Total time = 24.9469s    Reduced Chi^2 = 0.91458      Levenberg-Marquardt = 0.1
 Free parameter values:
          274          1.497          43.88             35       0.006967
 Total time = 25.848s    Reduced Chi^2 = 0.90934      Levenberg-Marquardt = 10
 Free parameter values:
        273.9          1.495          43.88             35       0.006987
 Total time = 26.7384s    Reduced Chi^2 = 0.90915      Levenberg-Marquardt = 1000
 Free parameter values:
        273.9          1.495          43.88             35       0.006988
 *** No improvement in chi-squared over previous iteration ***
 Total time = 26.7479s    Reduced Chi^2 = 0.88338
 Free parameter values:
        273.9          1.495          43.88             35       0.006988
--------------------------------------------------------------------------------
Fit converged:

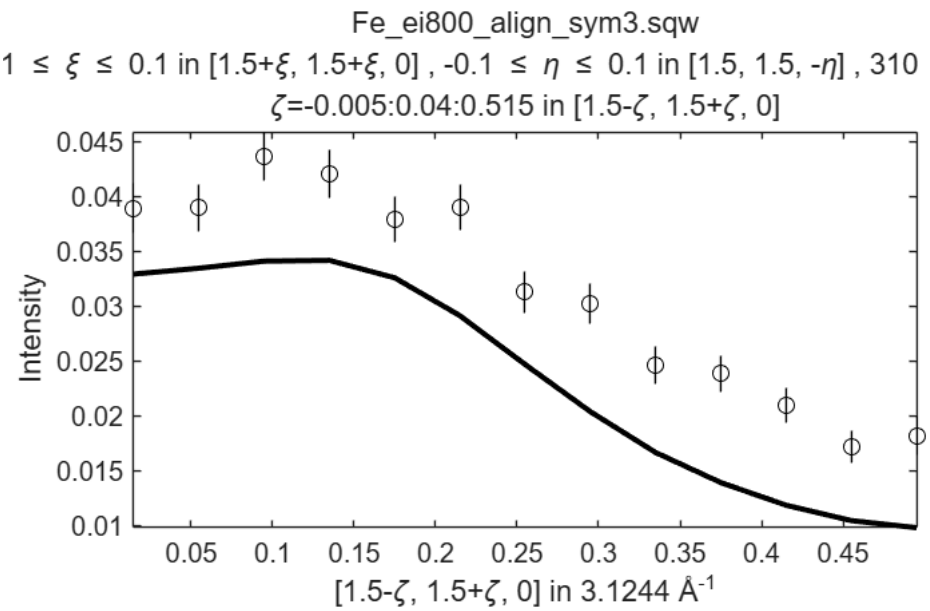
Parameter values (free parameters with error estimates):
---------------------------------------------------------

Foreground parameter values:
----------------------------
    1              1
    2              8
    3          273.9   +/-   200.6
    4          1.495   +/-   1.092
    5              0
    6          43.88   +/-   10.34
    7             35   +/-   34.89
    8              0
    9              0
   10              0


Background parameter values:
----------------------------
    1       0.006988   +/-   0.00599
    2              0
    3              0


Covariance matrix for free parameters:
--------------------------------------
    1.0000    0.9916    0.9895   -0.9380   -0.8810
    0.9916    1.0000    0.9665   -0.8885   -0.9325
    0.9895    0.9665    1.0000   -0.9741   -0.8114
   -0.9380   -0.8885   -0.9741    1.0000    0.6634
   -0.8810   -0.9325   -0.8114    0.6634    1.0000
```






```
*****************************
**** En = 325±5 going to 345
*****************************
-------------------------------------------------------------------------------
Beginning fit (max 20 iterations)-------------------------------------------------------------------------------------------
Starting point
------------------
 Total time = 0.76925s    Reduced Chi^2 = 106.0761
 Free parameter values:
         176.7         0.8006           40.8             35        0.02638
-------------------------------------------------------------------------------
Iteration = 1
------------------
 Total time = 4.7181s    Reduced Chi^2 = 2.6606      Levenberg-Marquardt = 0.1
 Free parameter values:
         161.1         0.6088          42.13             35        0.01567
-------------------------------------------------------------------------------
Iteration = 2
------------------
 Total time = 8.594s    Reduced Chi^2 = 2.0309      Levenberg-Marquardt = 0.01
 Free parameter values:
```

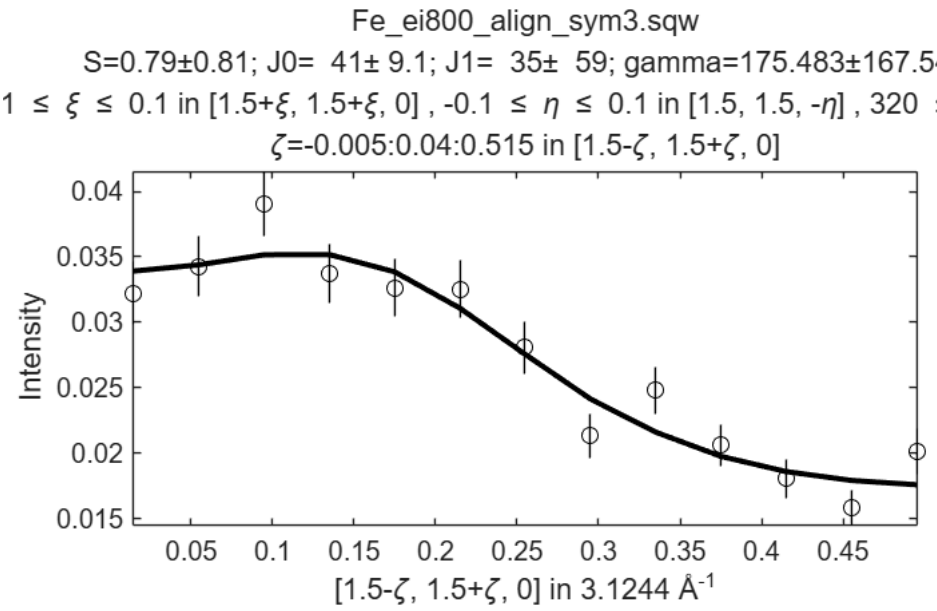
```
        156.9         0.6883          39.38             35        0.01269
-------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 12.5056s    Reduced Chi^2 = 1.9641      Levenberg-Marquardt = 0.001
 Free parameter values:
        172.3         0.7754          40.56             35        0.01179
-------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 16.4147s    Reduced Chi^2 = 1.9457      Levenberg-Marquardt = 0.0001
 Free parameter values:
        175.5         0.7928          40.75             35         0.0117
-------------------------------------------------------------------------------
Iteration = 5
------------------
 Total time = 20.1442s    Reduced Chi^2 = 1.9616      Levenberg-Marquardt = 1e-05
 Free parameter values:
        176.4         0.7977          40.78             35        0.01165
 Total time = 20.9179s    Reduced Chi^2 = 1.9666      Levenberg-Marquardt = 0.0001
 Free parameter values:
        176.4         0.7976          40.78             35        0.01166
 Total time = 21.7604s    Reduced Chi^2 = 1.9482      Levenberg-Marquardt = 0.001
 Free parameter values:
        176.2         0.7966          40.77             35        0.01166
 Total time = 22.5917s    Reduced Chi^2 = 1.9617      Levenberg-Marquardt = 0.1
 Free parameter values:
        175.6         0.7934          40.74             35        0.01169
 Total time = 23.4051s    Reduced Chi^2 = 1.9514      Levenberg-Marquardt = 10
 Free parameter values:
        175.5         0.7928          40.74             35         0.0117
 Total time = 24.1577s    Reduced Chi^2 = 1.9583      Levenberg-Marquardt = 1000
 Free parameter values:
        175.5         0.7928          40.75             35         0.0117
 *** No improvement in chi-squared over previous iteration ***
 Total time = 24.1661s    Reduced Chi^2 = 1.9457
 Free parameter values:
        175.5         0.7928          40.75             35         0.0117
-------------------------------------------------------------------------------
Fit converged:

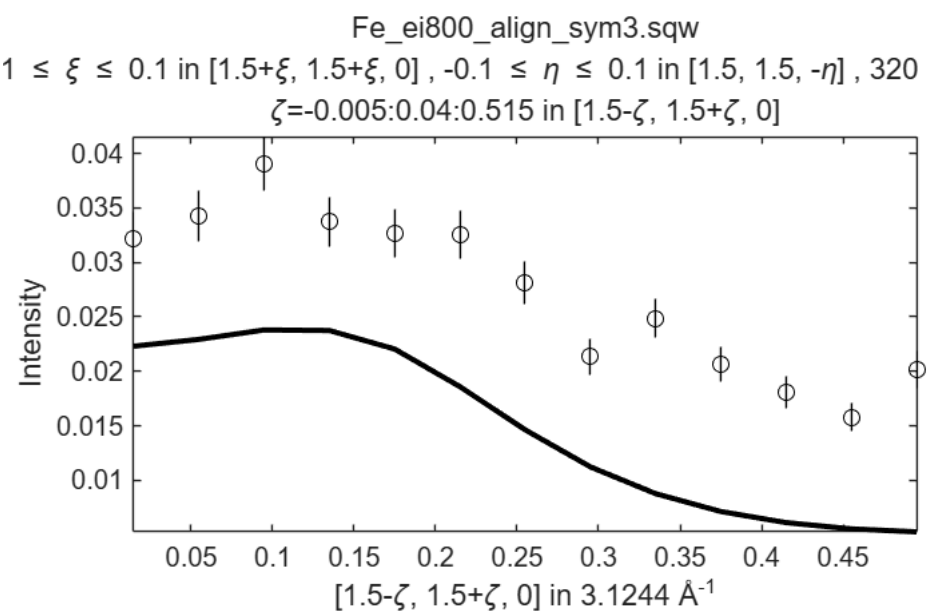
Parameter values (free parameters with error estimates):
---------------------------------------------------------

Foreground parameter values:
----------------------------
    1              1
    2              8
    3          175.5   +/-   167.5
    4         0.7928   +/-   0.8098
    5              0
    6          40.75   +/-   9.056
    7             35   +/-   59.23
    8              0
    9              0
   10              0


Background parameter values:
----------------------------
    1         0.0117   +/-   0.006355
    2              0
    3              0
```

```
Covariance matrix for free parameters:
-------------------------------------
    1.0000    0.9926    0.9761   -0.9953   -0.9245
    0.9926    1.0000    0.9566   -0.9827   -0.9568
    0.9761    0.9566    1.0000   -0.9908   -0.8382
   -0.9953   -0.9827   -0.9908    1.0000    0.8890
   -0.9245   -0.9568   -0.8382    0.8890    1.0000
```






```
*****************************
**** En = 335±5 going to 345
*****************************
--------------------------------------------------------------------------------
Beginning fit (max 20 iterations)----------------------------------------------------------------------------------------
Starting point
------------------
 Total time = 0.54249s    Reduced Chi^2 = 61.8803
 Free parameter values:
         178.2         0.7598          41.61             35        0.02516
--------------------------------------------------------------------------------
Iteration = 1
------------------
```

```
 Total time = 3.3904s    Reduced Chi^2 = 1.4621      Levenberg-Marquardt = 0.1
 Free parameter values:
        161.7         0.5828          42.85             35        0.01596
------------------------------------------------------------------------------
Iteration = 2
------------------
 Total time = 6.4593s    Reduced Chi^2 = 1.1046      Levenberg-Marquardt = 0.01
 Free parameter values:
        158.8         0.6587          40.21             35        0.01347
------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 9.4409s    Reduced Chi^2 = 1.079      Levenberg-Marquardt = 0.001
 Free parameter values:
          172         0.7297          41.24             35        0.01279
------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 12.4247s    Reduced Chi^2 = 1.0707      Levenberg-Marquardt = 0.0001
 Free parameter values:
        175.9         0.7486          41.49             35         0.0127
------------------------------------------------------------------------------
Iteration = 5
------------------
 Total time = 15.5409s    Reduced Chi^2 = 1.079      Levenberg-Marquardt = 1e-05
 Free parameter values:
        178.1           0.76           41.6             35        0.01262
 Total time = 16.1613s    Reduced Chi^2 = 1.0596      Levenberg-Marquardt = 0.0001
 Free parameter values:
        178.1         0.7597           41.6             35        0.01262
------------------------------------------------------------------------------
Iteration = 6
------------------
 Total time = 19.3598s    Reduced Chi^2 = 1.0761      Levenberg-Marquardt = 1e-05
 Free parameter values:
        175.9         0.7495          41.49             35        0.01269
 Total time = 19.9712s    Reduced Chi^2 = 1.0701      Levenberg-Marquardt = 0.0001
 Free parameter values:
          176         0.7498          41.49             35        0.01269
 Total time = 20.5909s    Reduced Chi^2 = 1.0681      Levenberg-Marquardt = 0.001
 Free parameter values:
        176.4         0.7518          41.52             35        0.01268
 Total time = 21.1771s    Reduced Chi^2 = 1.0697      Levenberg-Marquardt = 0.1
 Free parameter values:
        177.8         0.7583          41.59             35        0.01263
 Total time = 21.7877s    Reduced Chi^2 = 1.0722      Levenberg-Marquardt = 10
 Free parameter values:
        178.1         0.7595           41.6             35        0.01262
 Total time = 22.4422s    Reduced Chi^2 = 1.0753      Levenberg-Marquardt = 1000
 Free parameter values:
        178.1         0.7596           41.6             35        0.01262
 *** No improvement in chi-squared over previous iteration ***
 Total time = 22.4504s    Reduced Chi^2 = 1.0596
 Free parameter values:
        178.1         0.7597           41.6             35        0.01262
------------------------------------------------------------------------------
Fit converged:

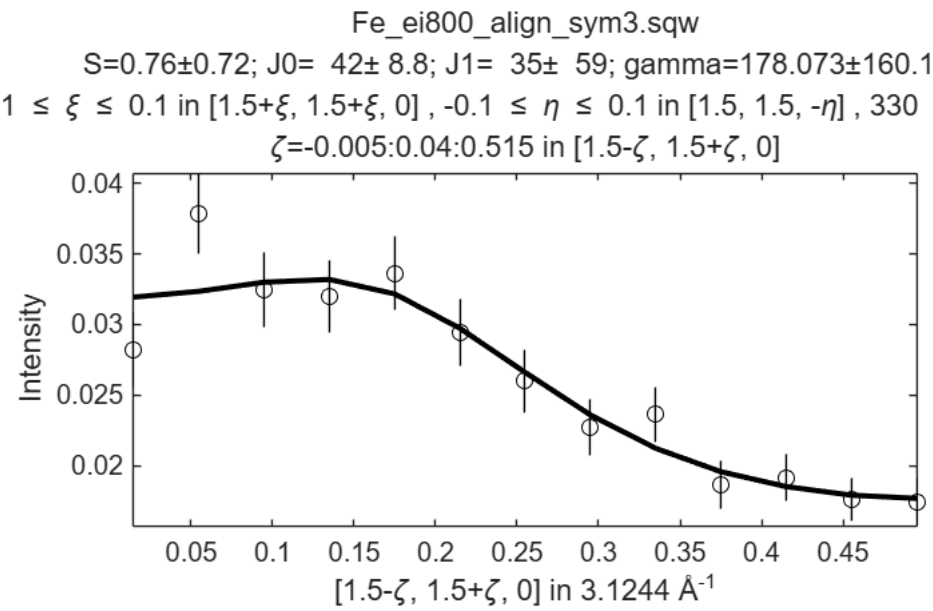
Parameter values (free parameters with error estimates):
---------------------------------------------------------

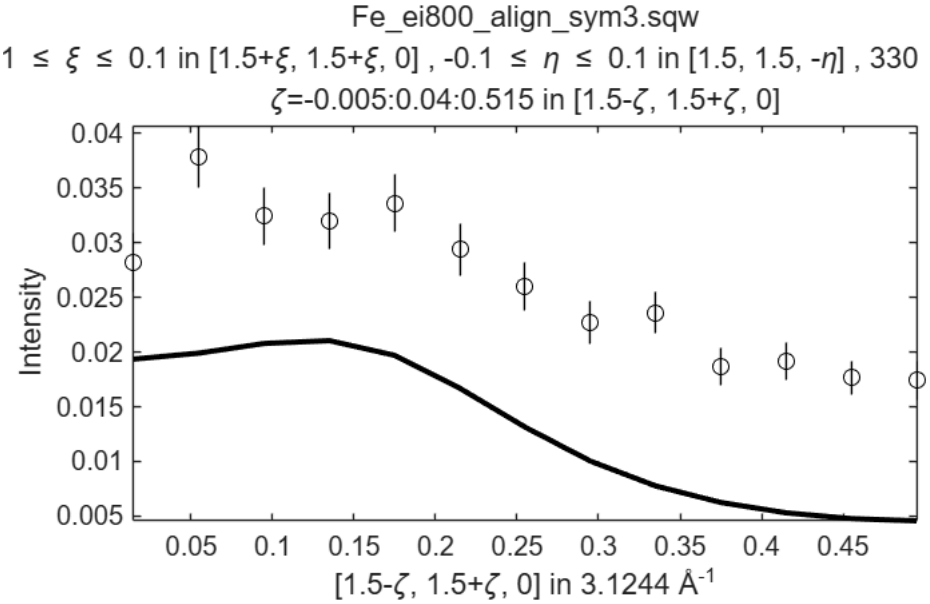
Foreground parameter values:
----------------------------
    1              1
```

```
    2              8
    3          178.1   +/-   160.1
    4         0.7597   +/-   0.718
    5              0
    6           41.6   +/-   8.758
    7             35   +/-   59.02
    8              0
    9              0
   10              0


Background parameter values:
----------------------------
    1        0.01262   +/-   0.004936
    2              0
    3              0


Covariance matrix for free parameters:
--------------------------------------
    1.0000    0.9925    0.9785   -0.9978   -0.9214
    0.9925    1.0000    0.9590   -0.9876   -0.9560
    0.9785    0.9590    1.0000   -0.9888   -0.8414
   -0.9978   -0.9876   -0.9888    1.0000    0.9002
   -0.9214   -0.9560   -0.8414    0.9002    1.0000
```






```
*****************************
**** En = 345±5 going to 345
*****************************
-------------------------------------------------------------------------------
Beginning fit (max 20 iterations)---------------------------------------------------------------------------------------------
Starting point
------------------
 Total time = 0.25262s    Reduced Chi^2 = 35.0837
 Free parameter values:
        182.3         0.9324          43.34             35        0.01895
-------------------------------------------------------------------------------
Iteration = 1
------------------
 Total time = 1.6751s    Reduced Chi^2 = 1.9894      Levenberg-Marquardt = 0.1
 Free parameter values:
        154.1         0.6964          44.28             35        0.01061
-------------------------------------------------------------------------------
Iteration = 2
------------------
 Total time = 3.0688s    Reduced Chi^2 = 1.77      Levenberg-Marquardt = 0.01
 Free parameter values:
        181.2         0.8839          43.09             35       0.007758
-------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 4.4057s    Reduced Chi^2 = 1.746      Levenberg-Marquardt = 0.001
 Free parameter values:
        175.6         0.9004          42.97             35       0.007396
-------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 5.7459s    Reduced Chi^2 = 1.7429      Levenberg-Marquardt = 0.0001
 Free parameter values:
        183.2          0.939           43.4             35       0.007193
-------------------------------------------------------------------------------
Iteration = 5
------------------
 Total time = 7.2124s    Reduced Chi^2 = 1.745      Levenberg-Marquardt = 1e-05
 Free parameter values:
        176.2          0.903          42.97             35       0.007363
 Total time = 7.4954s    Reduced Chi^2 = 1.746      Levenberg-Marquardt = 0.0001
 Free parameter values:
```

```
        176.4          0.904          42.98             35       0.007358
 Total time = 7.7979s    Reduced Chi^2 = 1.7513      Levenberg-Marquardt = 0.001
 Free parameter values:
        177.8         0.9109          43.06             35       0.007324
 Total time = 8.0573s    Reduced Chi^2 = 1.7365      Levenberg-Marquardt = 0.1
 Free parameter values:
        182.4         0.9347          43.33             35       0.007209
-------------------------------------------------------------------------------
Iteration = 6
-----------------
 Total time = 9.3629s    Reduced Chi^2 = 1.748      Levenberg-Marquardt = 0.01
 Free parameter values:
        181.6         0.9308          43.24             35       0.007204
 Total time = 9.6612s    Reduced Chi^2 = 1.7504      Levenberg-Marquardt = 0.1
 Free parameter values:
        182.2         0.9335          43.29             35       0.007198
 Total time = 9.9269s    Reduced Chi^2 = 1.754      Levenberg-Marquardt = 1
 Free parameter values:
        182.4         0.9343          43.31             35       0.007204
 Total time = 10.1958s    Reduced Chi^2 = 1.7435      Levenberg-Marquardt = 100
 Free parameter values:
        182.4         0.9347          43.33             35       0.007209
 Total time = 10.4562s    Reduced Chi^2 = 1.7539      Levenberg-Marquardt = 10000
 Free parameter values:
        182.4         0.9347          43.33             35       0.007209
 Total time = 10.7187s    Reduced Chi^2 = 1.7526      Levenberg-Marquardt = 1000000
 Free parameter values:
        182.4         0.9347          43.33             35       0.007209
 *** No improvement in chi-squared over previous iteration ***
 Total time = 10.7272s    Reduced Chi^2 = 1.7365
 Free parameter values:
        182.4         0.9347          43.33             35       0.007209
-------------------------------------------------------------------------------
Fit converged:

Parameter values (free parameters with error estimates):
---------------------------------------------------------

Foreground parameter values:
----------------------------
    1             1
    2             8
    3         182.4   +/-   306.6
    4        0.9347   +/-   1.611
    5             0
    6         43.33   +/-   18.02
    7            35   +/-   95.17
    8             0
    9             0
   10             0


Background parameter values:
----------------------------
    1       0.007209   +/-   0.008586
    2             0
    3             0


Covariance matrix for free parameters:
--------------------------------------
    1.0000    0.9916    0.9802   -0.9896   -0.9298
    0.9916    1.0000    0.9656   -0.9759   -0.9567
    0.9802    0.9656    1.0000   -0.9964   -0.8557
```

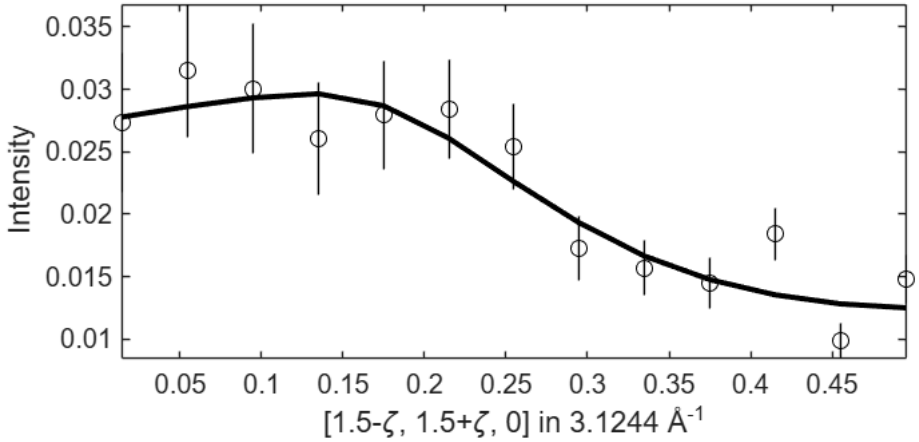
```
   -0.9896   -0.9759   -0.9964    1.0000    0.8745
   -0.9298   -0.9567   -0.8557    0.8745    1.0000
```

Fe_ei800_align_sym3.sqw

S=0.93± 1.6; J0= 43± 18; J1= 35± 95; gamma=182.45±306.583

1 ≤ ξ ≤ 0.1 in [1.5+ξ, 1.5+ξ, 0] , -0.1 ≤ η ≤ 0.1 in [1.5, 1.5, -η] , 340 ≤ E ≤

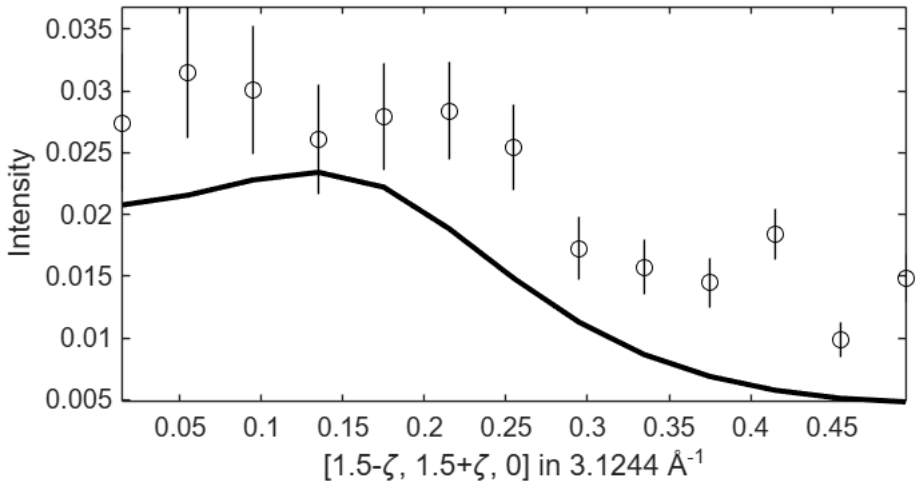
ζ=-0.005:0.04:0.515 in [1.5-ζ, 1.5+ζ, 0]

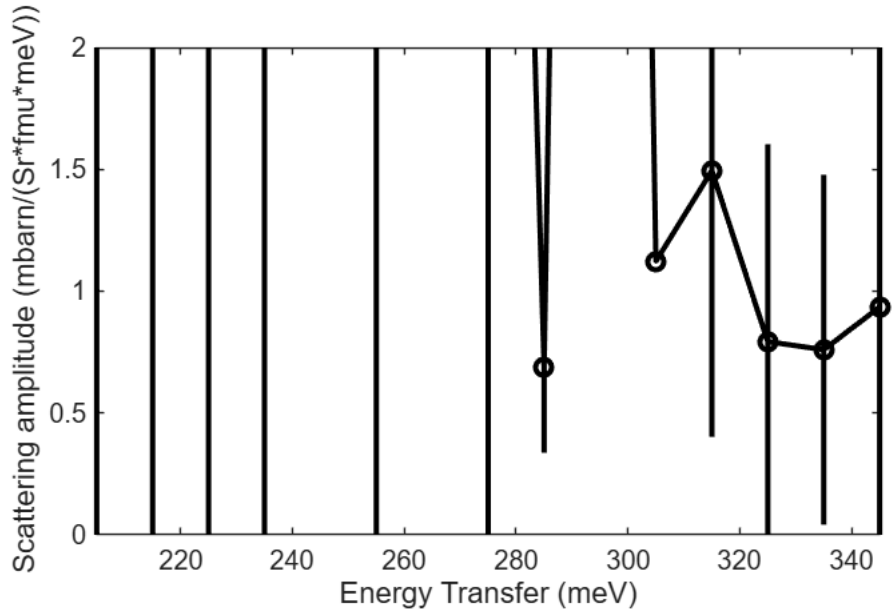


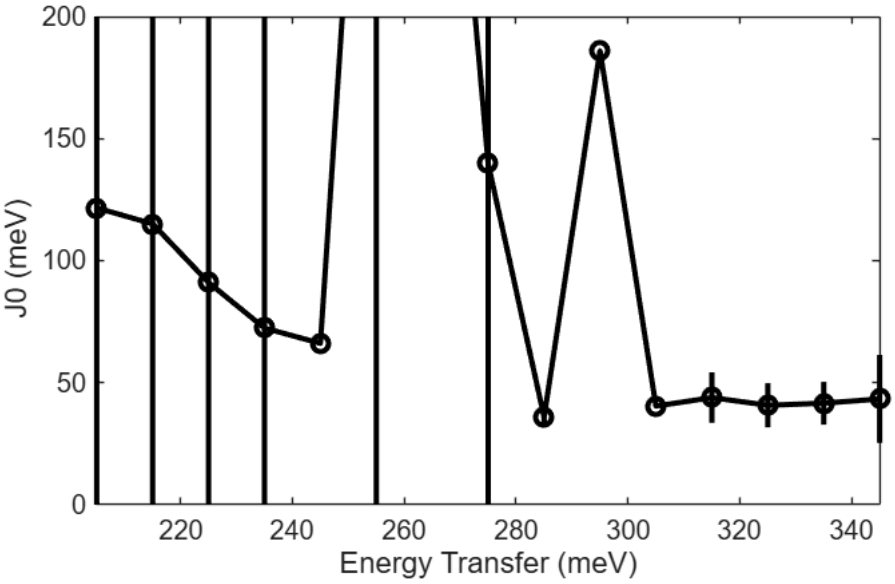
Fe_ei800_align_sym3.sqw

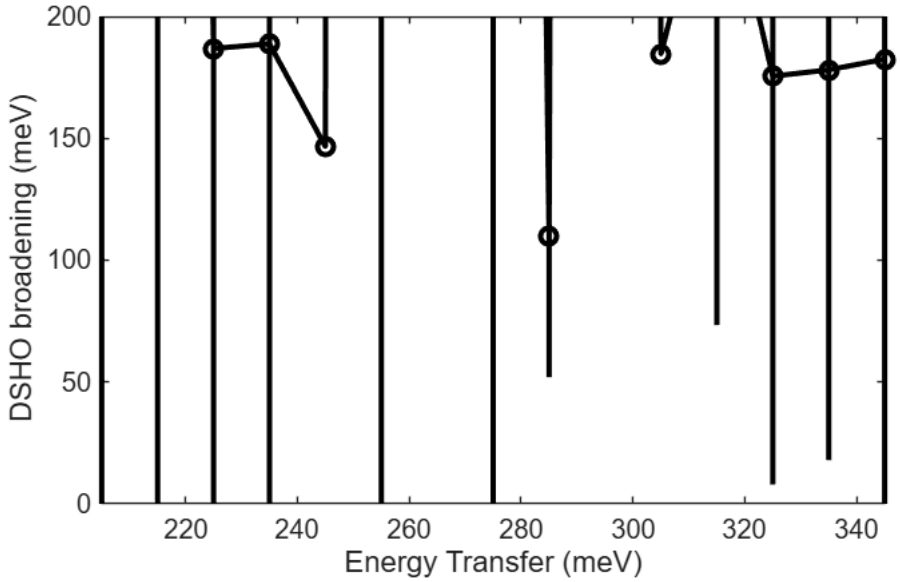
1 ≤ ξ ≤ 0.1 in [1.5+ξ, 1.5+ξ, 0] , -0.1 ≤ η ≤ 0.1 in [1.5, 1.5, -η] , 340 ≤ E ≤

ζ=-0.005:0.04:0.515 in [1.5-ζ, 1.5+ζ, 0]




Scattering amplitude (mbarn/(Sr*fmu*meV))
2
1.5
1
0.5
0
220
240
260
280
300
320
340
Energy Transfer (meV)

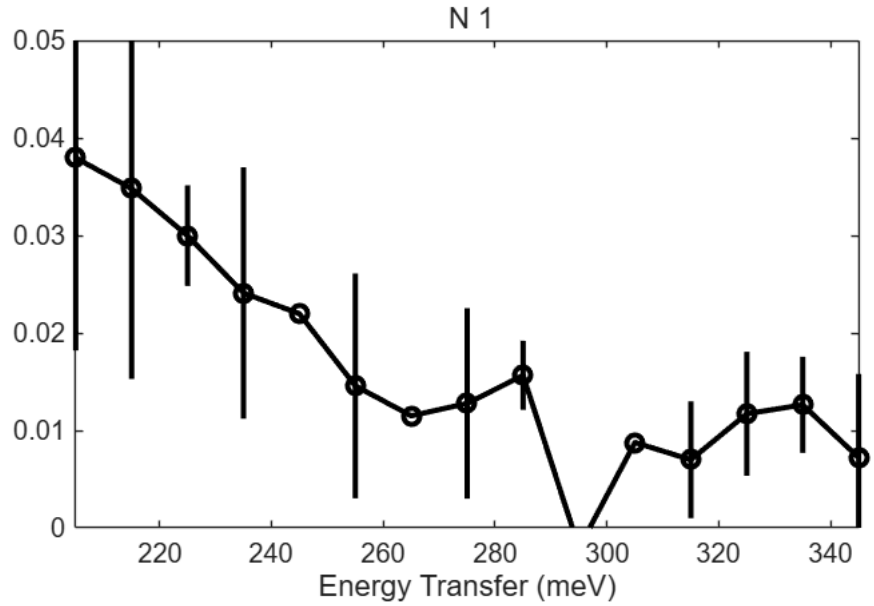

J0 (meV)
200
150
100
50
0
220
240
260
280
300
320
340
Energy Transfer (meV)



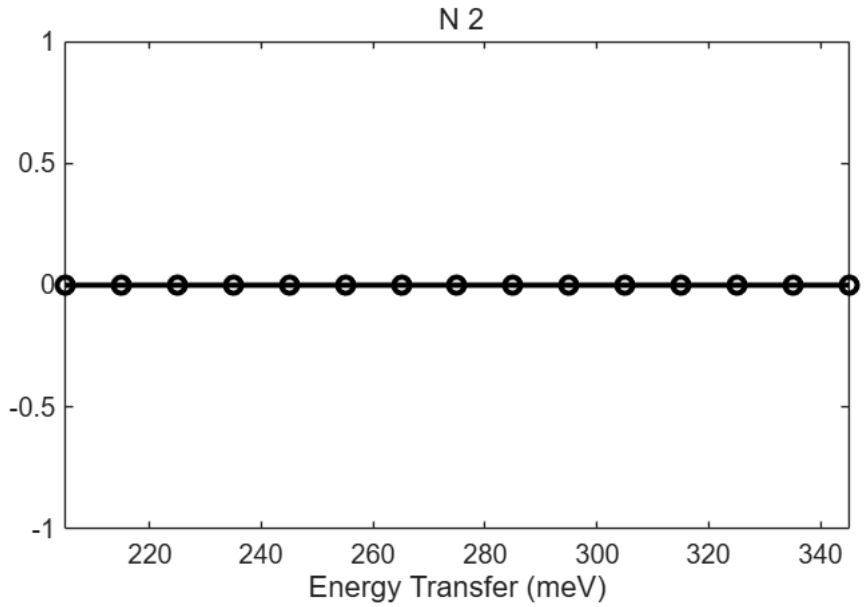

```
cl = HN_FFCutsFits.(selectCuts);
if iscell(cl)
    HN_FFCutsFits.(selectCuts) = [cl{:}];
end

beep
beep
beep
```

```
all_fit_par = HN_FFCutsFits.([selectCuts,'_2Jng_fit_par']);
bg_data = extract_bg_par(all_fit_par,[1,2]);
% original slope parameters
pd(bg_data(1));lx(all_fit_par(1).en,all_fit_par(end).en);ly 0 0.05; keep_figure;
```



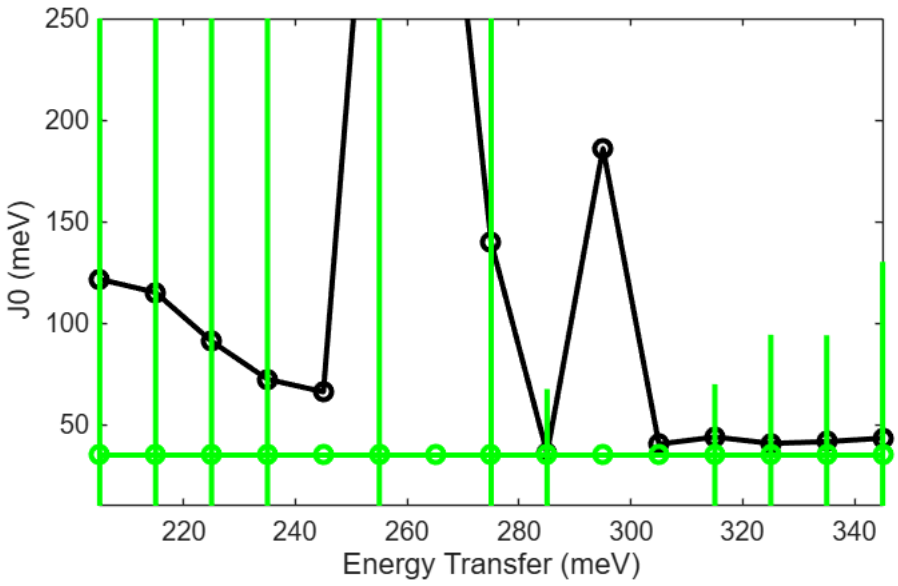

```
pd(bg_data(2));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure
```




```
%pd(bg_data(3));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_fi
gure;
%pd(bg_data(4));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_fi
gure
```

```
doc = matlab.desktop.editor.getActive;
doc.save;
thisFile = 'e:\SHARE\Fe\Data\analysis\06_fit_with_AQ0\fit_cutFe_HN.m';
targ_file = fullfile(fileparts(thisFile),['fit_cutFe_HN_',selectCuts,'.pdf']);
export(thisFile,targ_file);
%close all;
%end
```

```
J0 = extract_fit_par(all_fit_par,6,'J0','meV');
J1 = extract_fit_par(all_fit_par,7,'J1','meV');
acolor k; pd(J0);
acolor g; pd(J1);ly 10 250;keep_figure;
```




```
fit_src_struc = HN_FFCutsFits;
save(CutFitFileHN,"fit_src_struc",'-v7.3')
```